Onshore Drilling Well-Pad Design

Jakarta Drilling Seciety

Soreang, 16 June 2022

Well-Pad Design List Of Content



- Drilling Rig Model and WellPad
- General Drilling Rig Activity
- Drilling Rig Equipment
- Existing WellPad
- New WellPad Design
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  - WellPad Sizing
  - Conductor Design
  - Cellar Design
  - WellPad Design

DATI 253 Rigging Up-1

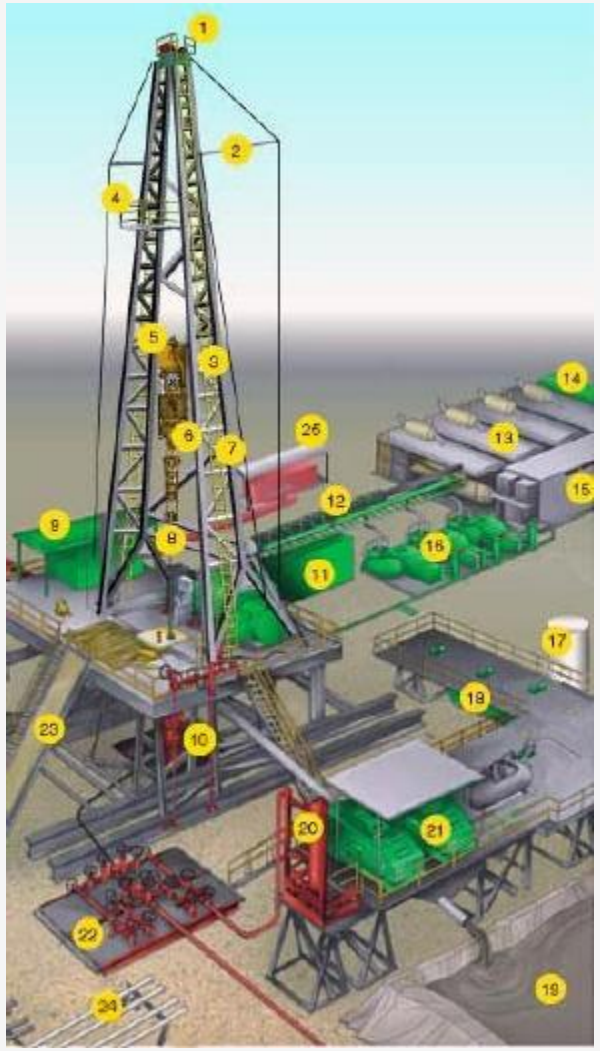
DATI 253 Rigging Up-2

Discussion Objectives

- Understanding basic Wellpad design to meet drilling operation requirement, HSE aspect and effective drilling cost.
- To have detail drilling concerns list related to Wellpad as a basis for discussion with Civil Construction team during design phase.
- In future short term plan JDS able to build Standard Operating Procedure related drilling rig Wellpad.

INTRODUCTION

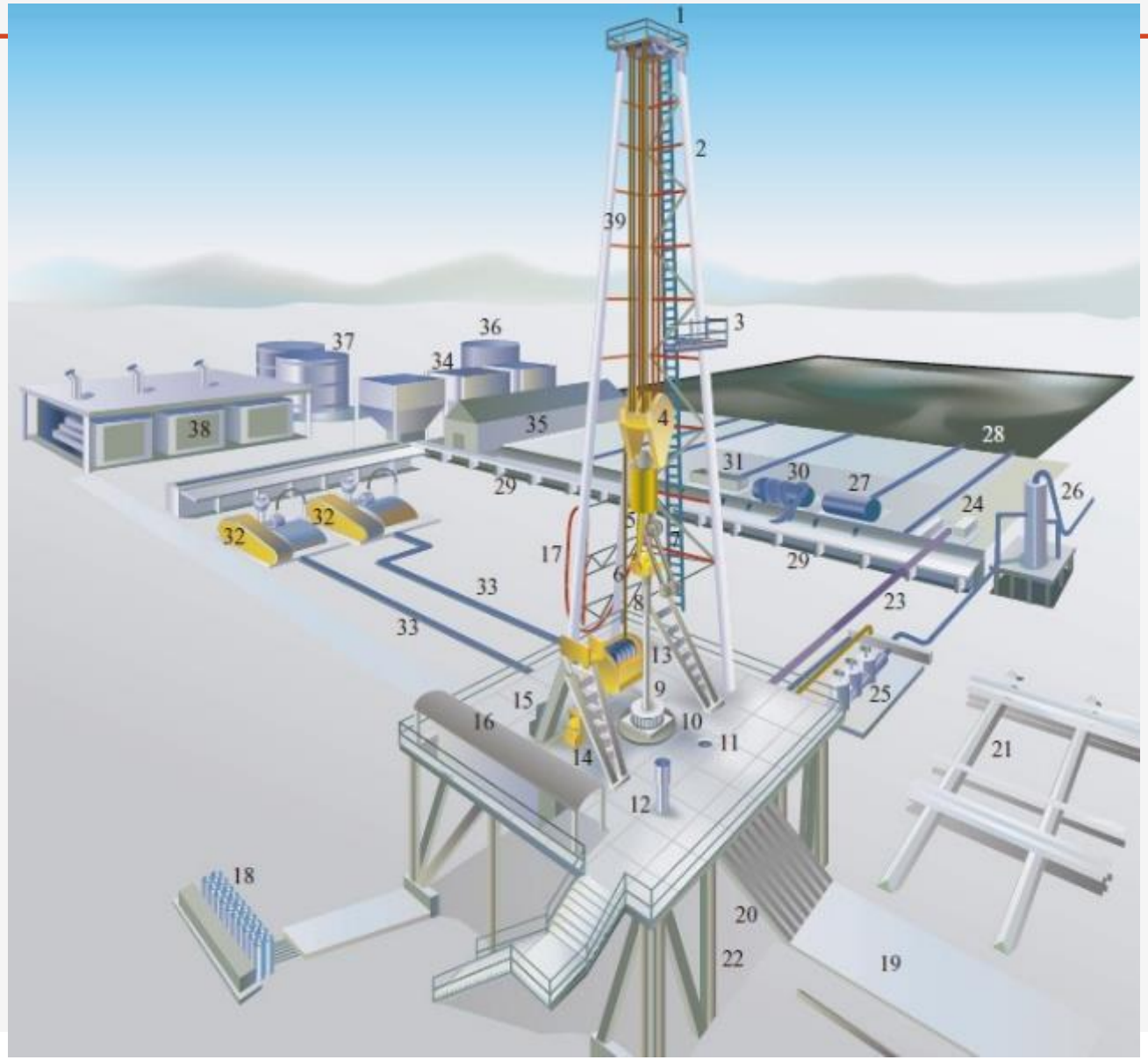
PERALATAN PEMBORAN



- Derrick Mast
- Top Drive System
- Drawwork
- Drill Floor
- Dog House
- Blow Out Preventer
- Pumps
- Solid Control Equipment
- Genset
- Warehouse
- Camps

Main Rig Component

1 crown block
2 mast
3 monkey board
4 traveling block
5 hook
6 swivel
7 elevators
8 kelly
9 kelly bushing
10 master bushing
11 mousehole
12 rathole
13 drawworks
14 weight indicator
15 driller's console
16 doghouse
17 rotary hose
18 accumulator unit
19 catwalk
20 pipe ramp



21 pipe rack
22 substructure
23 mud return line
24 shale shaker
25 choke manifold
26 mud gas separator
27 degasser
28 reserve pit
29 mud pits
30 desander
31 desilter
32 mud pumps
33 mud discharge lines
34 bulk mud components storage
35 mud house
36 water tank
37 fuel storage
38 engines and generators
39 drilling line

Rig Moving – Truck & Trailer


Low Boy


Winch Truck


Vacuum Truck


Hi Boy


Dump Truck

Rig Moving – Alat Berat













RIGING UP



Raised Derrick Proses of TRUCK MOUNTED Rig

CEMENTING EQUIPMENT

Cementing Pumping Unit





Cementing Cilo



Batch Mixer



WECO 1502 CHIKSAN





HYDRAULIC FRACTURING



DRILLING INTRODUCTION



- Land Rig
- Offshore – Shallow Water
- Offshore – Deep Water
- Swamp Barge Rig

Oil and Gas Drilling Work Location













Geothermal Drilling Work Location











Geothermal Wellsite view



Karaha Bodas



Wayang Windu Well-Pad and Piping







KS Orka – Sorik Marapi







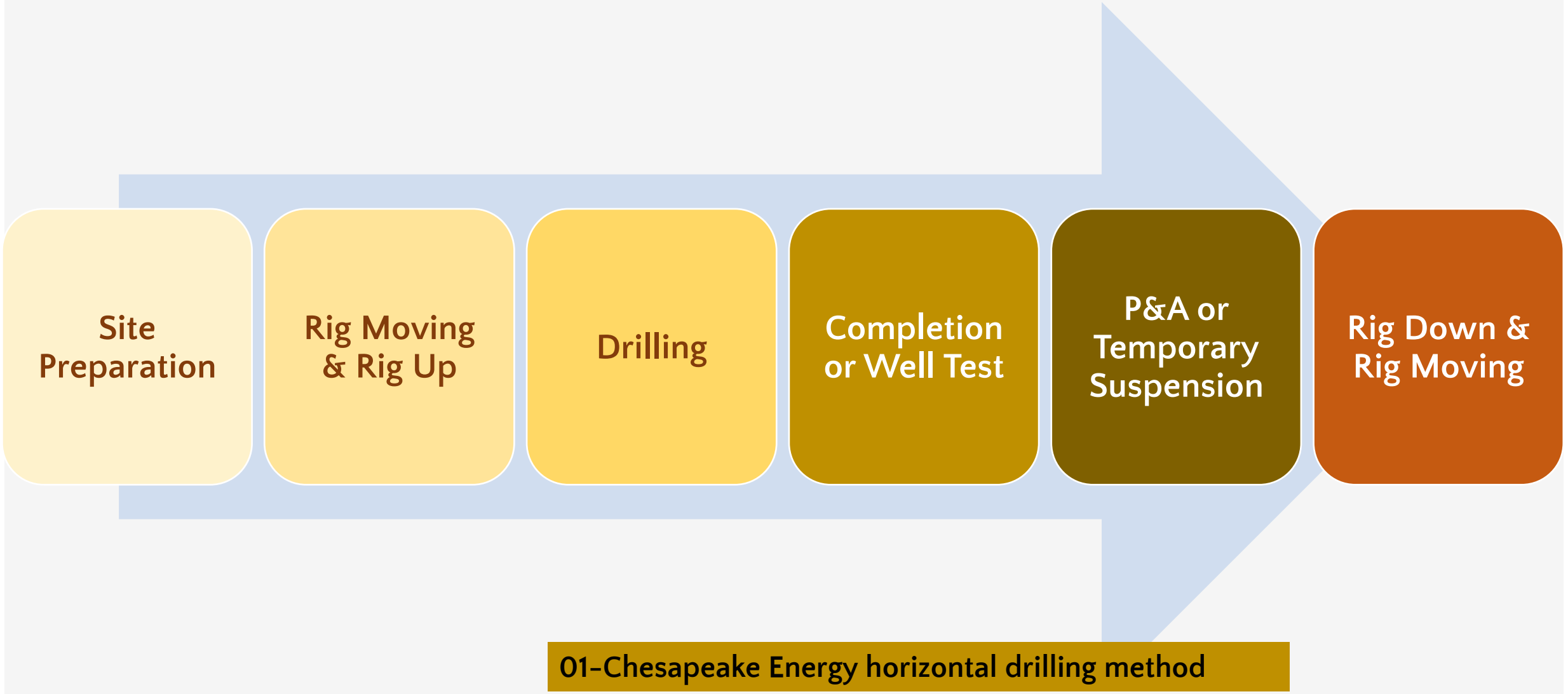
KS-Orka – Sokoria







Drilling Operation General Sequences



TYPICAL ONSHORE DRILLING ACTIVITIES

Pre-Drilling

- WellPad Construction
- Rig Mobilization
- Rig Moving
- Rigging Up

Drilling

- Spud the well
- Drilling Operation
- Set Casing dan Cementing
- Coring Operation
- Electric Logging
- Well Completion
- Well Test
- Plug and Abandonment
- Rigging Down
- Rig Demobilization

Post Drilling

- Well Stimulation
- Workover Operation

DRILLING RISK



RIG MOVING RISK









Why 

Oklahoma Oilfield Drilling Rig Accident



Landslide















COMMUNITY ISSUE













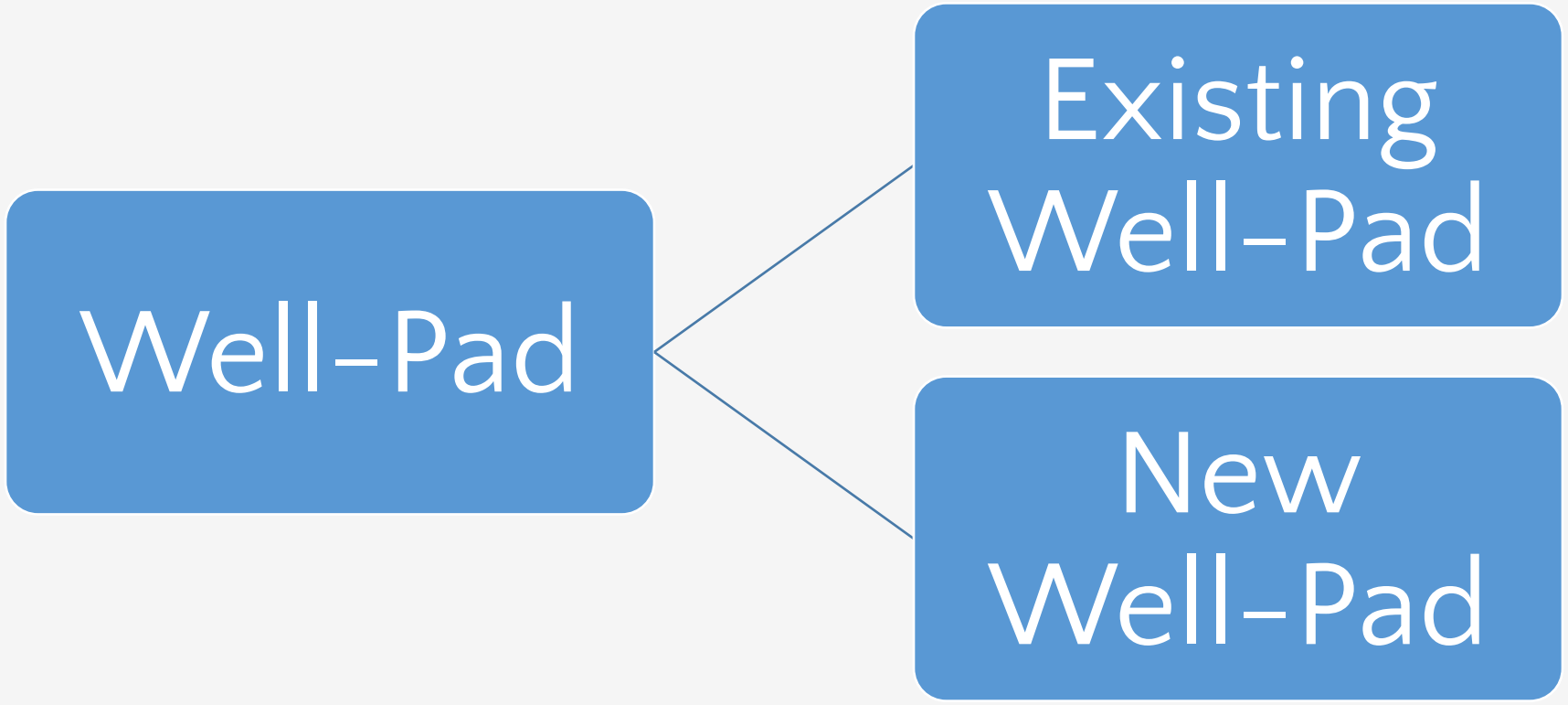




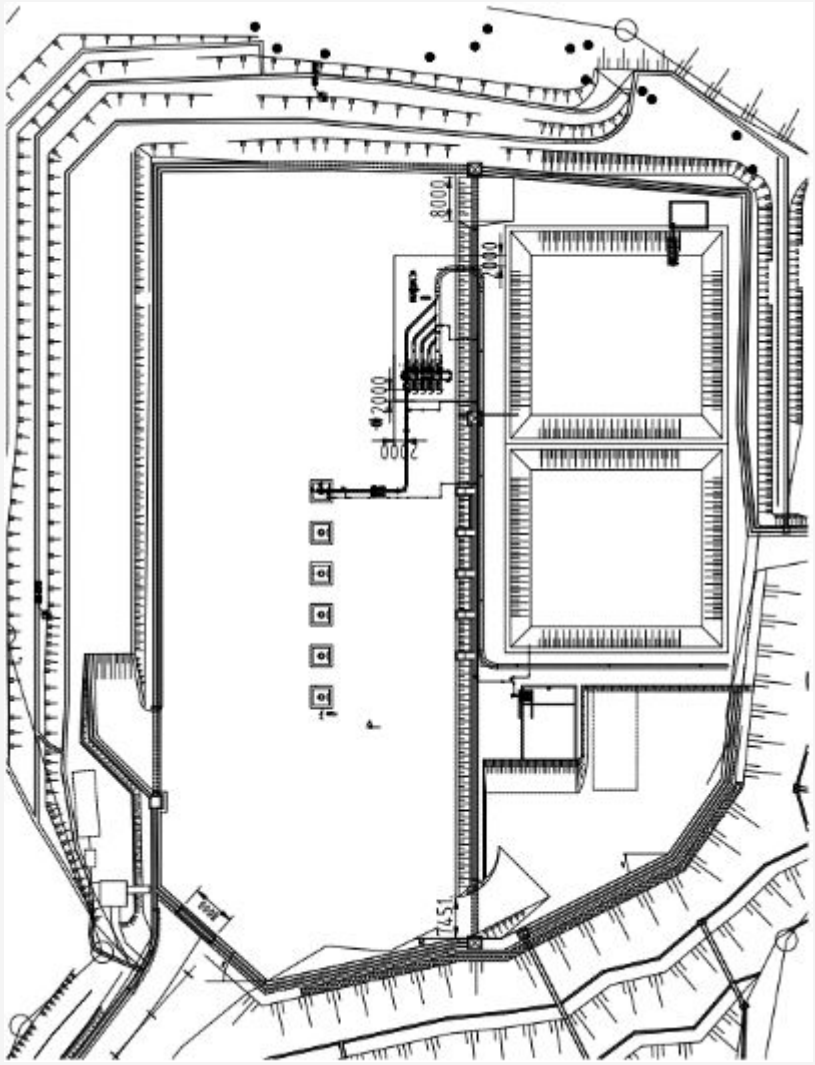


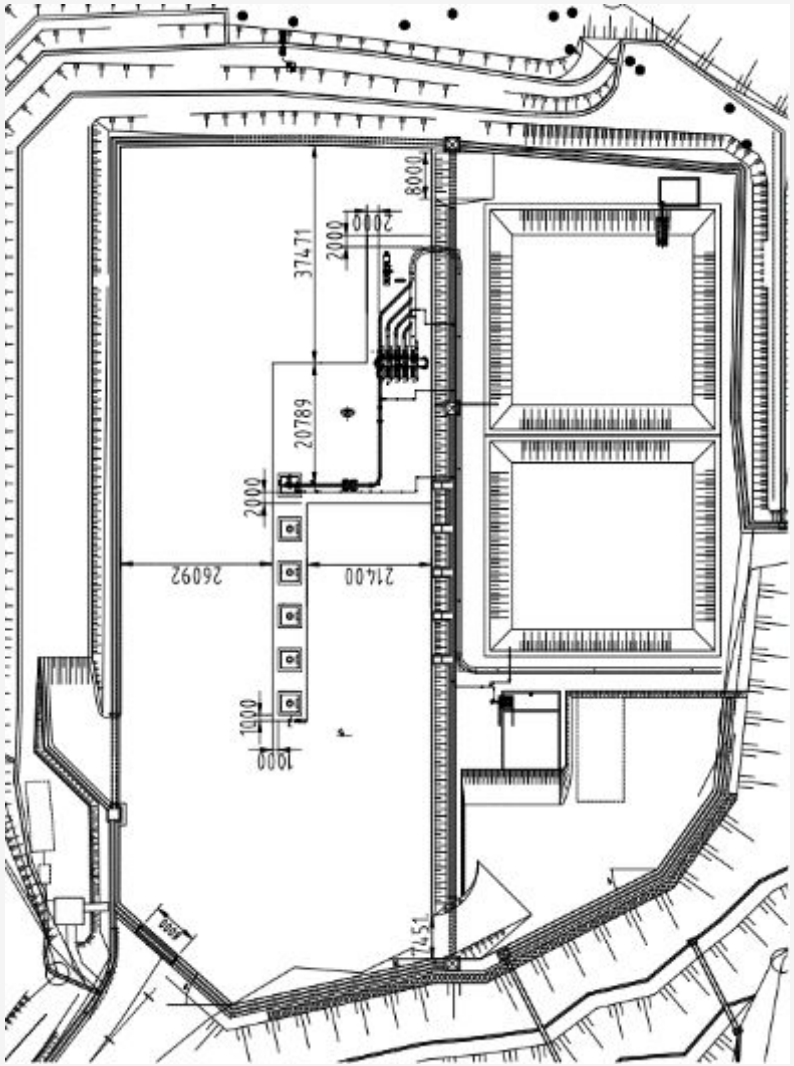
DRILLING in EXISTING WELLPAD

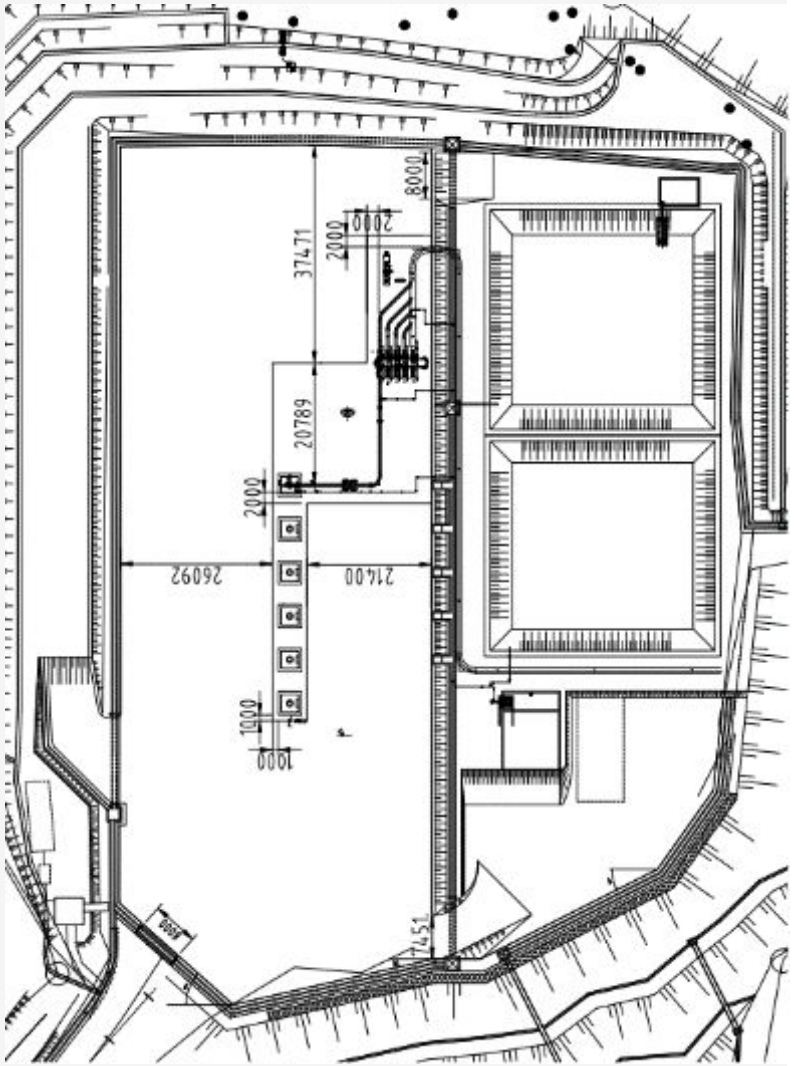
Well-Pad Design



Existing Well-Pad: Preparation Stage

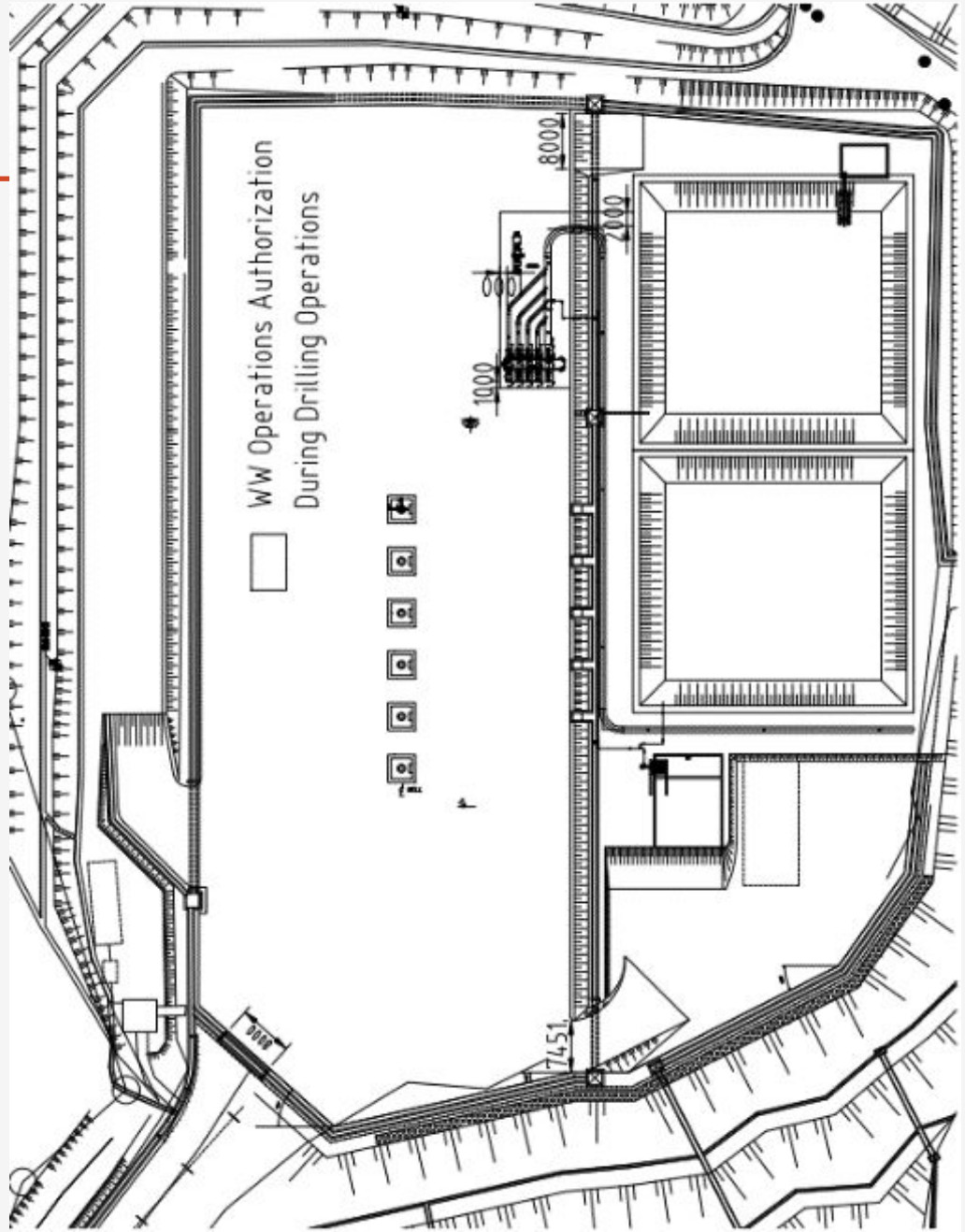




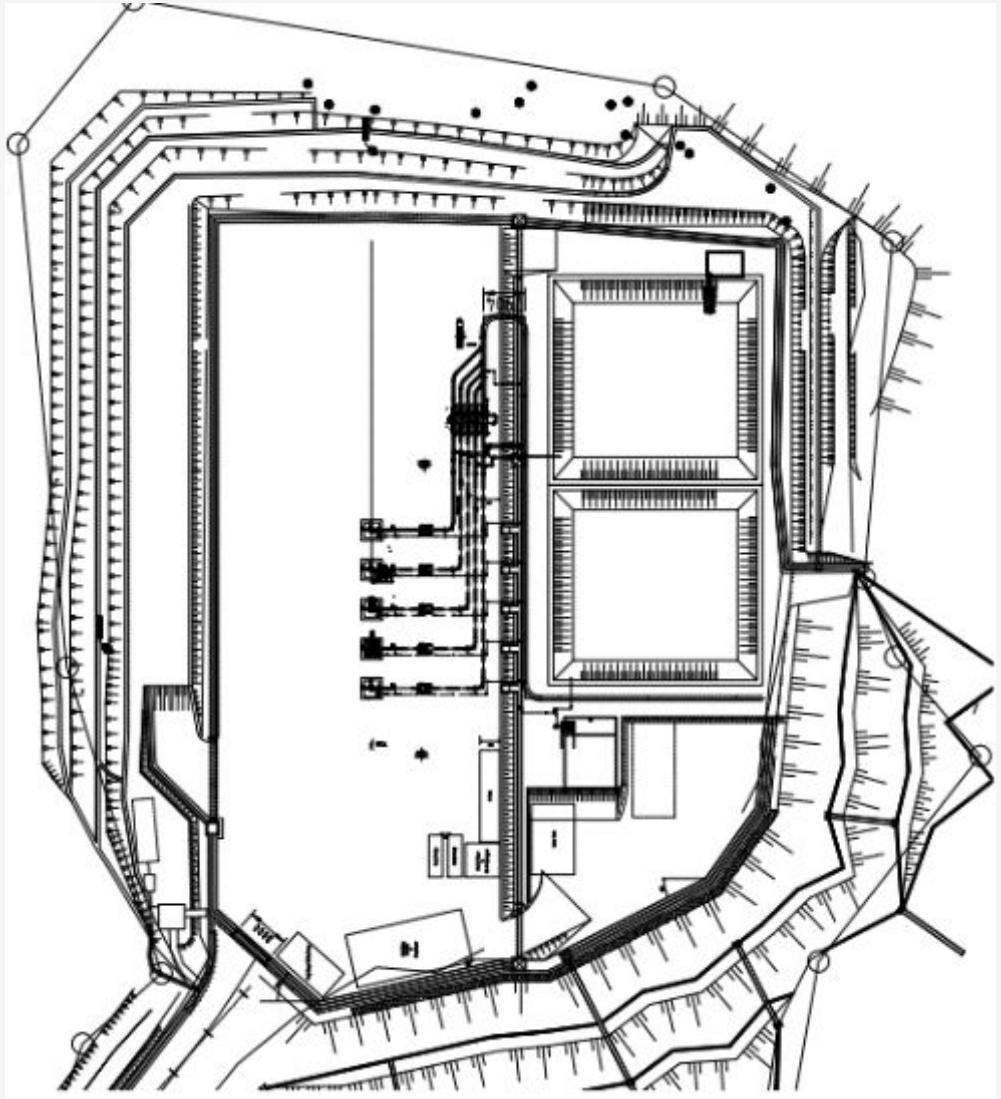


Existing Well-Pad: During Drilling

- Detail Well-Pad measurement
- Cek elevation
- Cek Drainage system
- Existing wellhead Protection
- Dismantle obstruction: pipes and others
- Simultaneous Operation (SIMOP) document
- Existing Septic Tanks
- Communication line
- Electric Cable Layout
- Control Panel Cable Layout
- Muster Point
- Digging permit
- Water and Mud Pond clean out or upgraded
- Soil Bearing Load Test
- Fencing surrounding WellPad
- Noise Level
- New Community close with Well-Pad
- Water Pipe Line: New and or exiting
- Water Pump Pad
- **Optimize RIG LAYOUT**



Existing Well-Pad: Post Drilling



- SIMOP:
  - Well Test
  - Production Piping Installation
- **Workover RIG LAYOUT capability**
- **Wellpad Reclamation**
- **Mud Pond Cleanout**

DRILLING in NEW WELLPAD

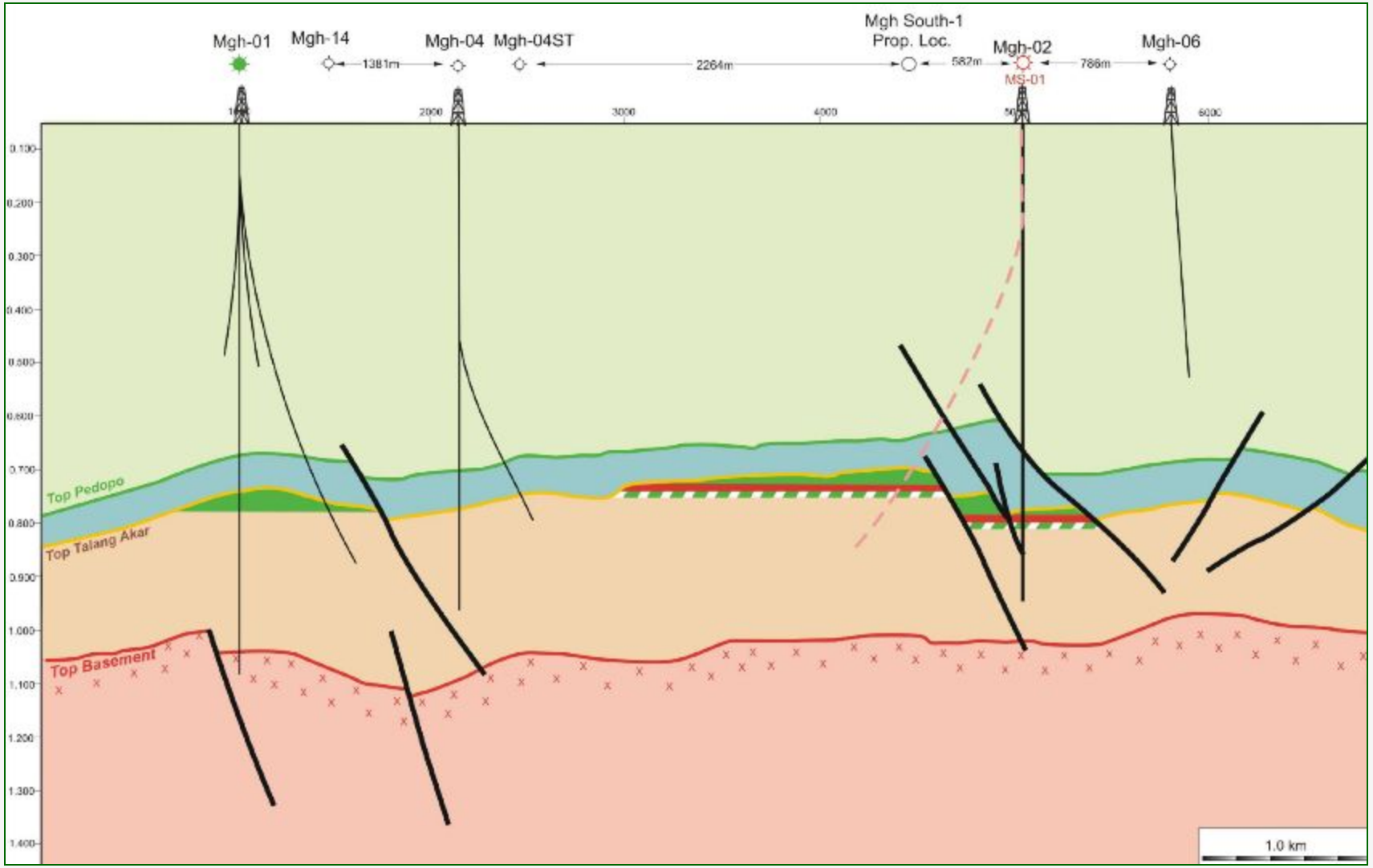
WellPad Refer to **SNI 13-6910-2002**

- Wells should be located at least **100 meters away from Public Roads**, Railways, Public Works, Houses or other places win which a source of ignition may arise.
- Vent, flare pits and extremities of flare lines should be located at least **90 meters from railways, roads, public works, processing units or tanks**. They should be at least **30 meters from a well**, gas/oil separator, site drainage or other possible source of ignitable vapors.
- Combustible material where possible all tress, undergrowth, and other such combustible material should be cleared from drilling sites and for a distance of **50 meters around the site perimeter**.
- Full earthing shall be provided, verified and documented. The maximum permissible resistance to earth shall be **4 ohms**. Where necessary boreholes to provide a ground to earth will have to be provided.

Potential Accident Risk and Concerns

- Fall Object
- Heavy Equipment crash
- Oil Spill
- Steam Exposure
- Noise Level
- Pollution
- H2S Release
- Burning
- Pipe leaking
- High Pressure line
- Vibration while drilling
- Collision Risk
- Community Issue
- Demonstration
- Land slide
- Flooding
- Water Pond Leaking
- Cold Temperature
- Raining Season
- Dry Season – Water supply
- Permits and Regulation
- Resources

DEVELOPMENT WELLS

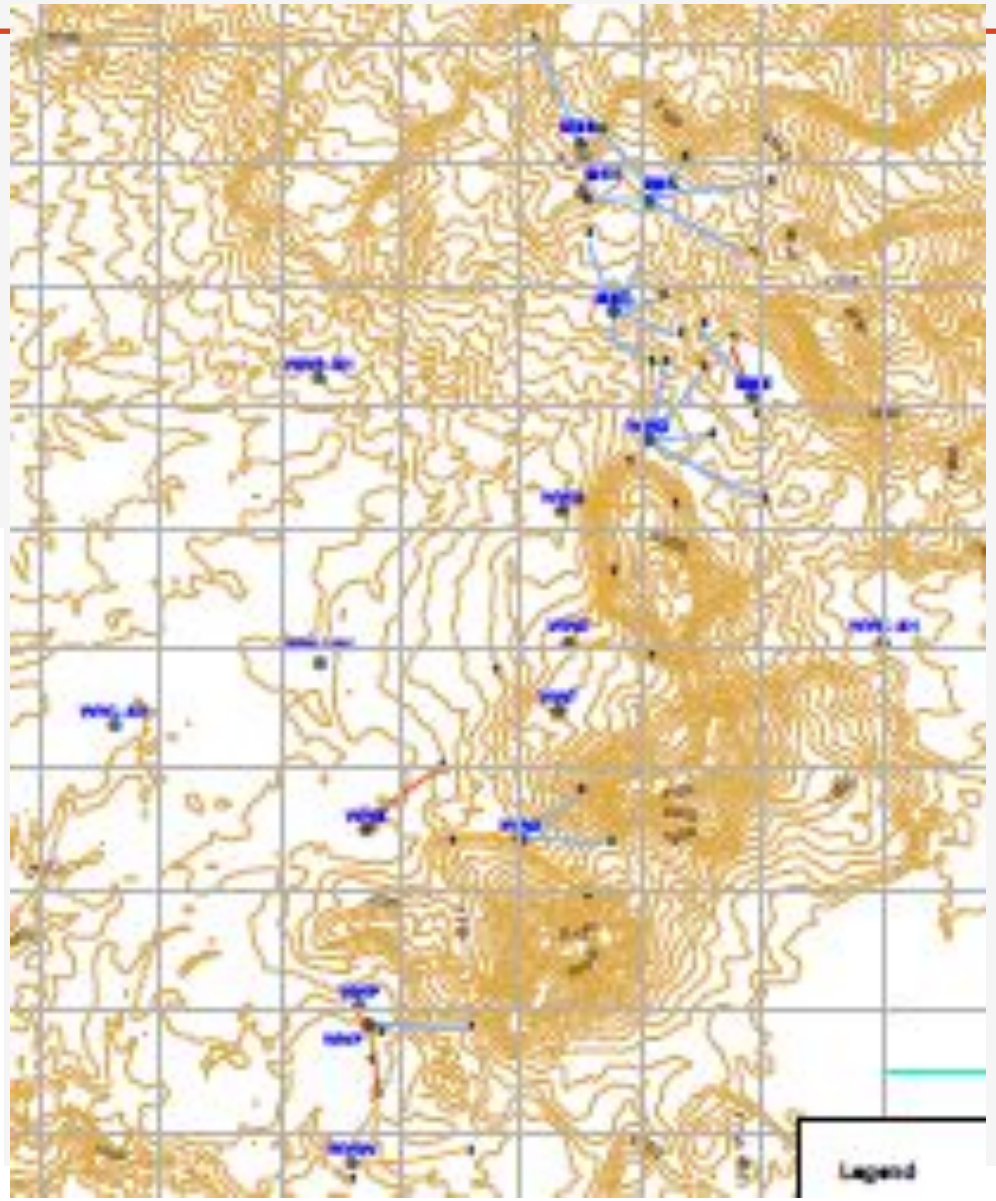


WAYANG WINDU GEOTHERMAL

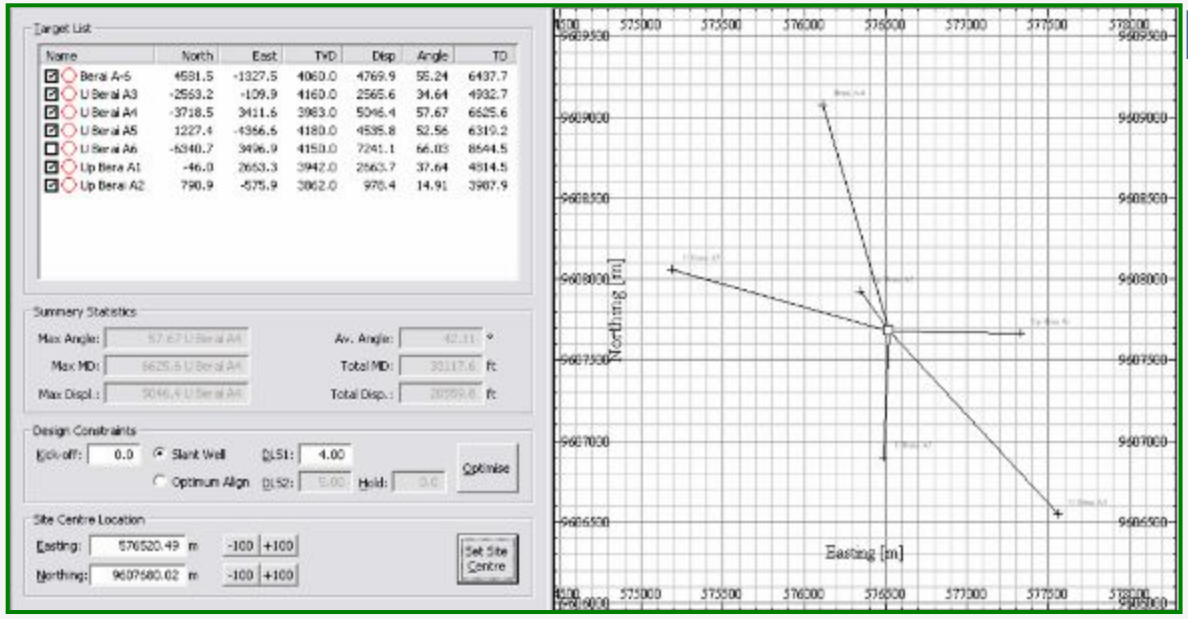


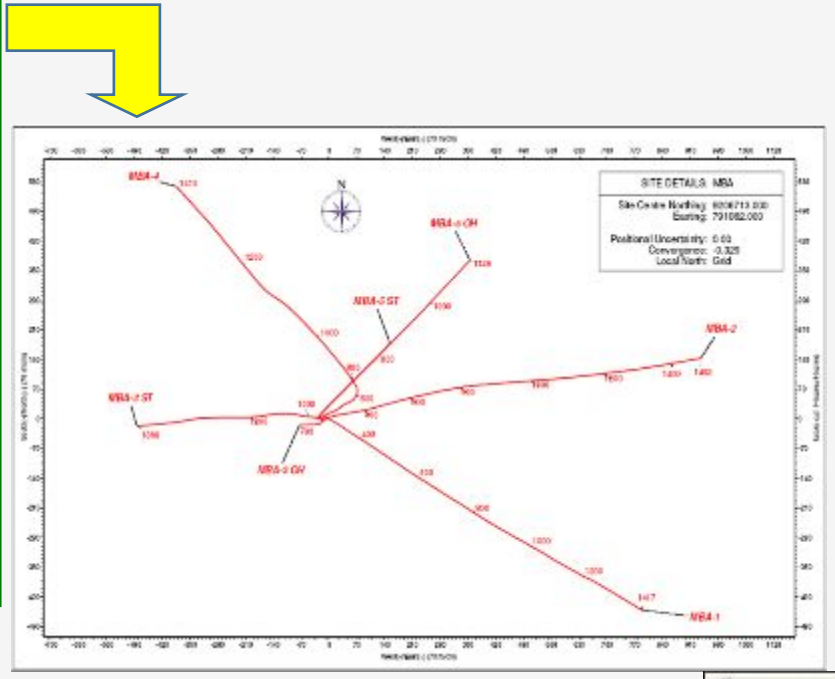




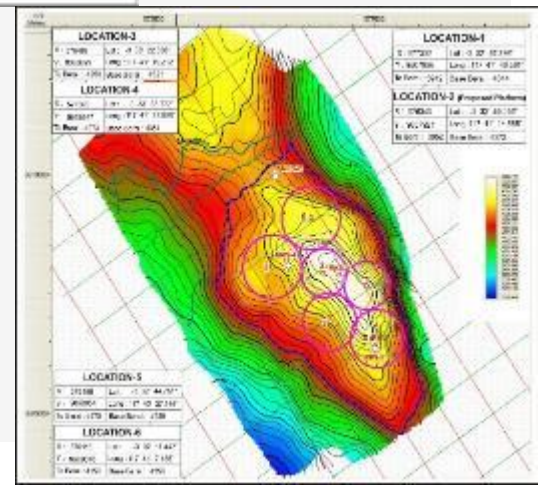


Optimum Surface Location

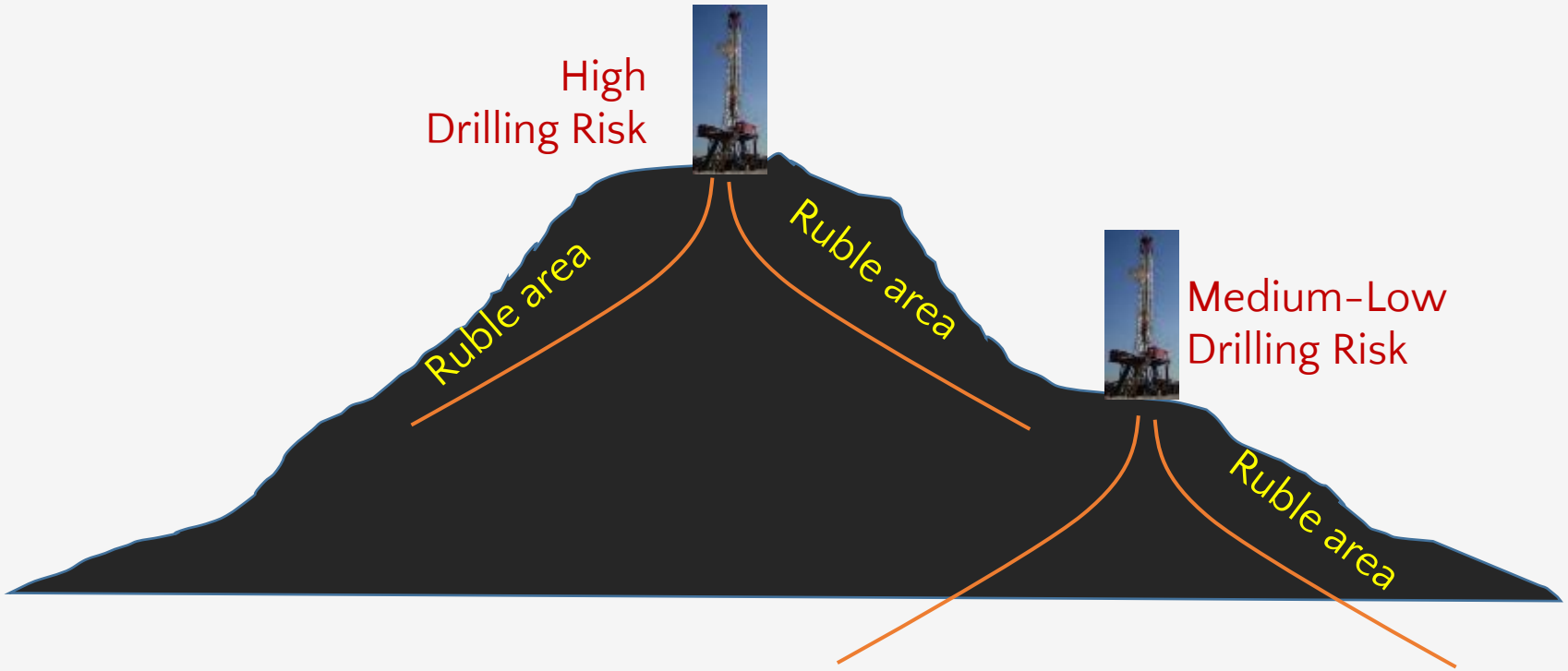




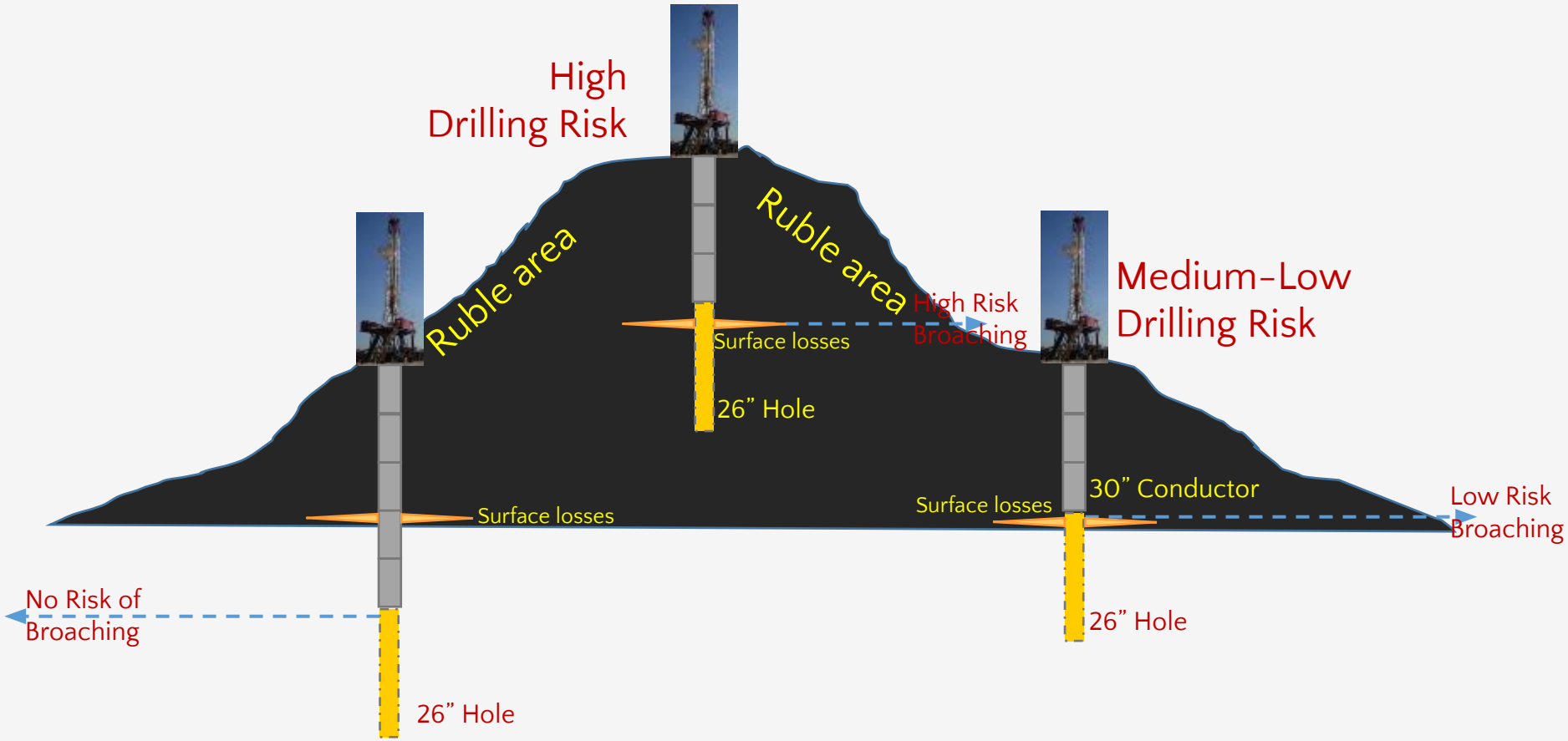
- Optimize Surface coordinate – software
- Site Survey
- Rig Layout and Service Company Layout
- Water Supply and Pipeline
- Cellar Design, Distance between Cellar, Well Collision Risk



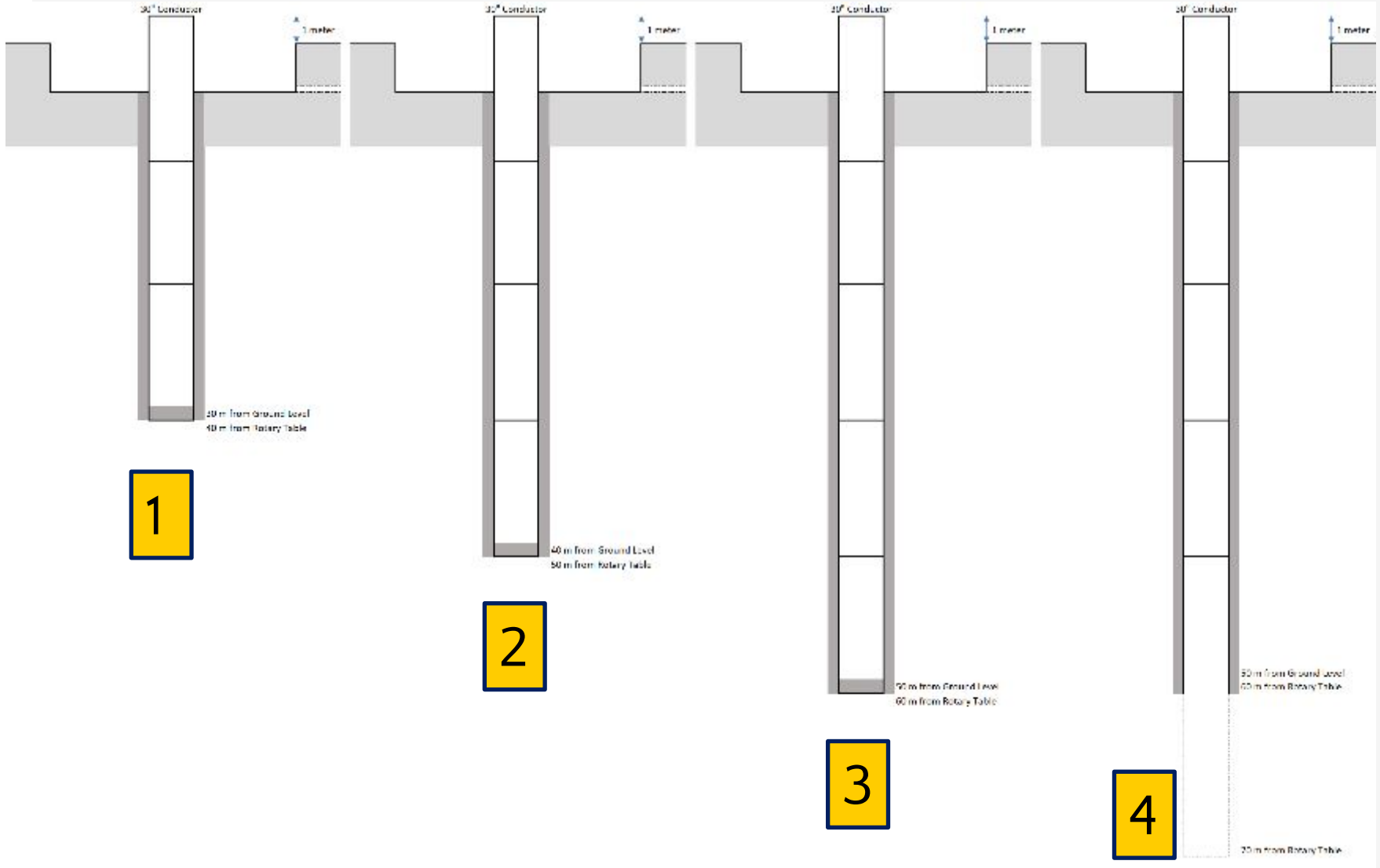
WellPad Surface Location



30” Conductor Depth



Conductor Depth Illustration



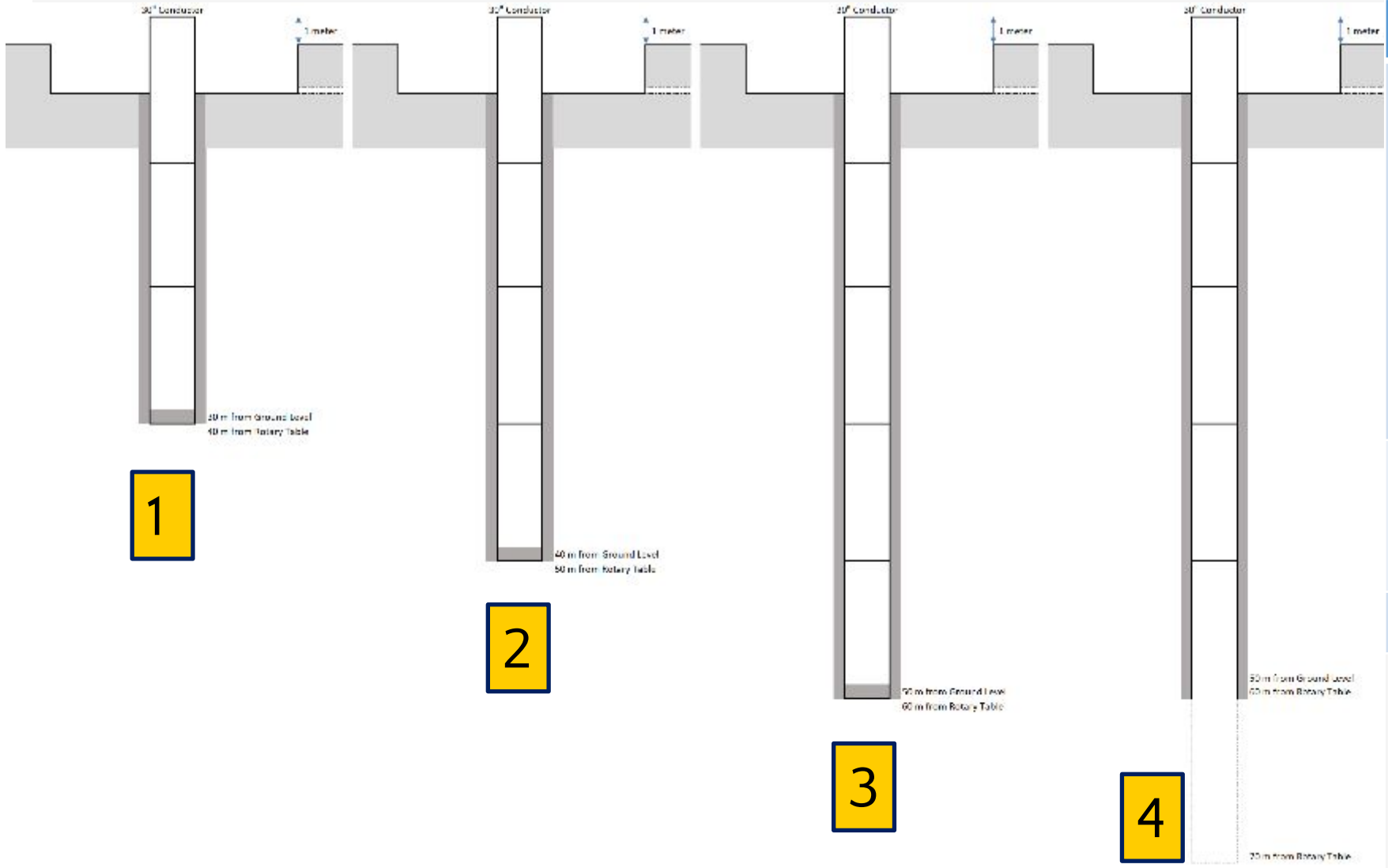
26" Directional BHA:

9-1/2" MM + 9-1/2" EM MWD Tool

No	BHA Component	OD	ID	Connection			LENGTH	Cumm
		in	in	Up	Down	Conn.	m	Length
1	26" Bit	26		TFA	7⅝ Reg		P	0.5	0.5
2	9-5/8" Mud Motor	9⅝	4.75	7⅝ Reg	7⅝ Reg	B/B	10.7	11.2
3	9 1/2" Catcher Sub w/ Float Sub	9½	3	7⅝ Reg	7⅝ Reg	B/P	1	12.2
4	24" IBS Stabilizer	9½	3	7⅝ Reg	7⅝ Reg	B/P	1.5	13.7
5	Shock Sub 9 1/2"	9½	3	7⅝ Reg	7⅝ Reg	B/P	2	15.7
6	9-1/2" NMDC	9½	3	7⅝ Reg	7⅝ Reg	B/P	9.1	24.8
7	9-1/2" MWD XEM 800	9½	3	7⅝ Reg	7⅝ Reg	B/P	9.1	33.9
8	9-1/2" MWD Gap Sub	9½	3	7⅝ Reg	7⅝ Reg	B/P	9.1	43
9	9-1/2" NMDC (Optional)	9½	3	7⅝ Reg	7⅝ Reg	B/P	9.1	52.1
10	9-1/2" Pony NMDC (Optional)	9½	3	7⅝ Reg	7⅝ Reg	B/P	3.0	55.1
11	X/O (7-5/8" REG Pin x 6 5/8" REG)	9½	3	6⅝ Reg	7⅝ Reg	B/P	3.0	58.1
12	7 x 8" DC	8	2.812	6⅝ Reg	6⅝ Reg	B/P	63.7	121.8
13	8" Hydraulic Jars	8	2.812	6⅝ Reg	6⅝ Reg	B/P	10.7	132.5
14	8" DC	8	2.812	6⅝ Reg	6⅝ Reg	B/P	9.1	141.6
15	X/O 6 5/8" RP x 4 1/2" IF9	8	2.812	4½ IF	6⅝ Reg	B/P	3.0	144.6
16	6 3/4" DC	6¾	2.813	4½ IF	4½ IF	B/P	9.1	153.7
17	15 x 5" HWDP	5	3	4½ IF	4½ IF	B/P	136.5	290.2
18								

- 26” Directional BHA length = > 58 m
- Option-3 and 4, we could run 26” Directional BHA without any drill out rotary BHA
- Rig time saving = 20 hours

Conductor Depth Illustration



Option-1 & 2	Option-3 & 4
MU Rotary BHA RIH Drill to 70 m POOH	20 hours
MU DD BHA RIH Drill to TD POOH	MU DD BHA RIH Drill to TD POOH
Additional cost for Auger and Casing =	Rig time saving = 20 hours USD 58 K
Potential Cost Saving:	

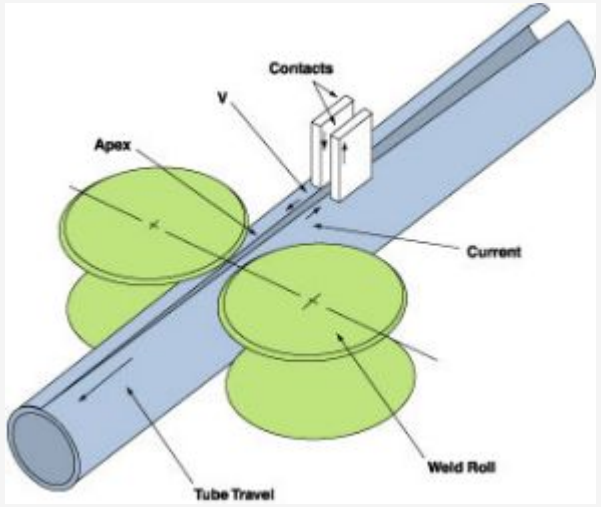
Advantages to set Deeper 30” Conductor Depth

- Rig time saving since it could run directional BHA instead of initiated with Rotary BHA to provide sufficient space for Directional BHA

- If any surface losses, has been cured during Augering operation which much less operating daily rate.

However, it need to confirm what the maximum depth that Auger could drill.

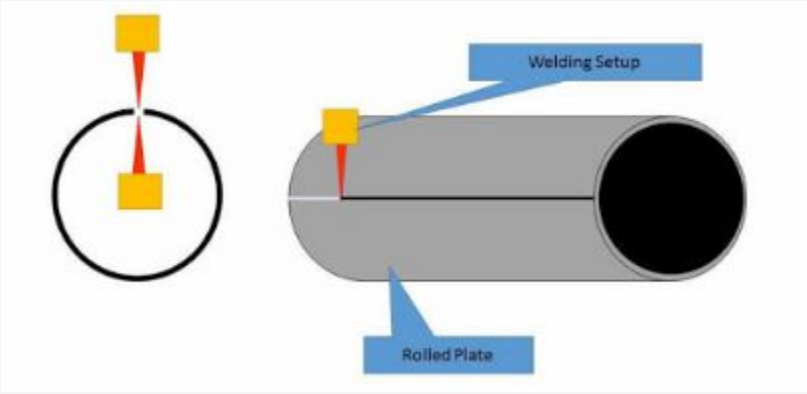
30” CASING CONDUCTOR MANUFACTURE PROCESS



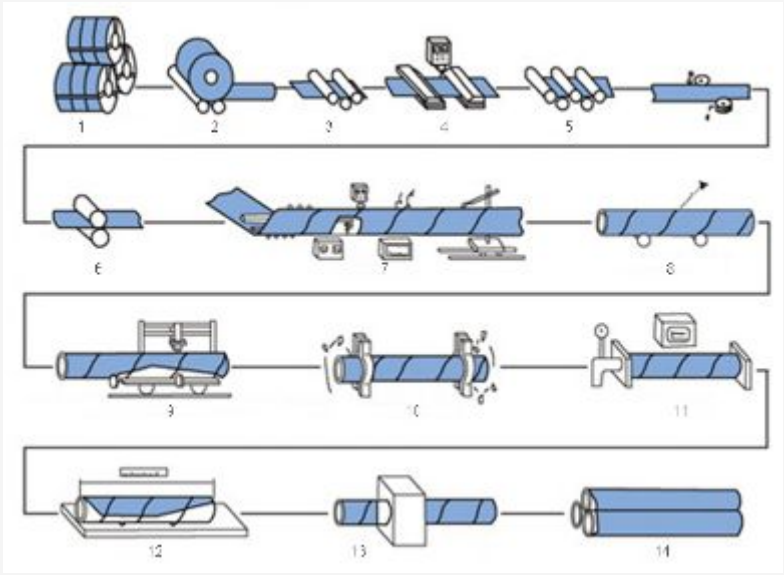
ERW: Electric Resistance Welded



SAW: Submerge Arc Welding



LSAW: Longitudinal Submerge Arc Welding
DSAW: LSAW with double seam weld in inside and outside



Spiral SAW

CASING MANUFACTURE PROCESS

ERW: Electric Resistance Welded

- Manufactured by rolling metal and then welding it longitudinally across its length

SAW: Submerge Arc Welding

LSAW: Longitudinal Submerge Arc Welding,

- Single seam weld on the outer surface

DSAW: LSAW with double seam weld in inside and outside

SPIRAL SAW



SEAMLESS:

- Manufactured by extruding the metal to the desired length. 20-30% higher price than ERW casing

30” Conductor Price Comparison

30” Casing	Welded	USD/Jt
1” Wall Thickness (For Piling)	ERW	
	LSAW	
	DSAW	
	SPIRAL	
	Seamless	

Stronger pipe, for piling installation



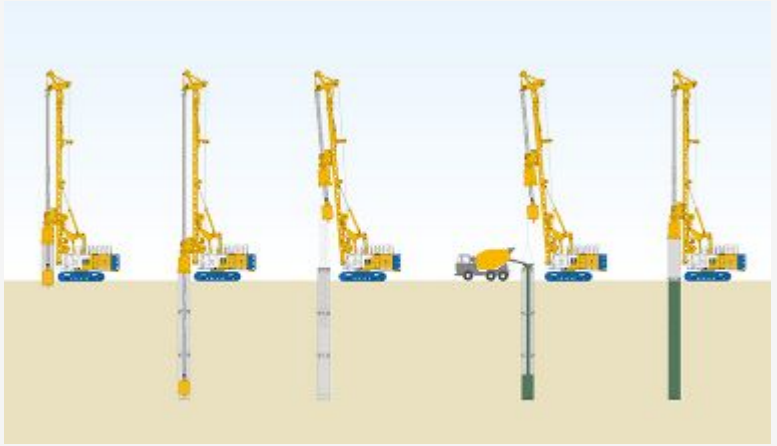
30” Casing	Welded	USD/Jt
0.75” Wall Thickness (Recommended for Auger)	ERW	
	LSAW	
	DSAW	
	SPIRAL	
	Seamless	

Thin pipe, for Auger installation



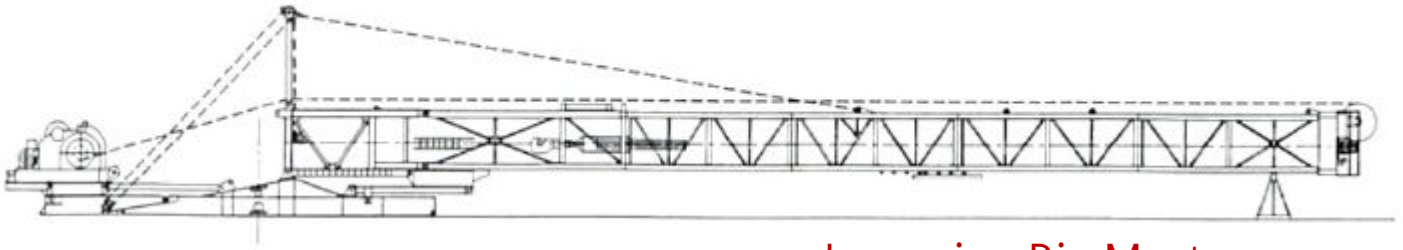
30” Casing	Welded	USD/Jt
0.5” Wall Thickness (Low Price)	ERW	
	LSAW	
	DSAW	
	SPIRAL	
	Seamless	

Thin pipe, for Auger installation



Detail Rig Layout





Lowering Rig Mast

Major Rig Equipment:

- Rig Sub Structure.
- Mud Tanks and Mud Pumps
- Rig Power Unit
- Rig Mini Camps & Containers

Major Service Company:

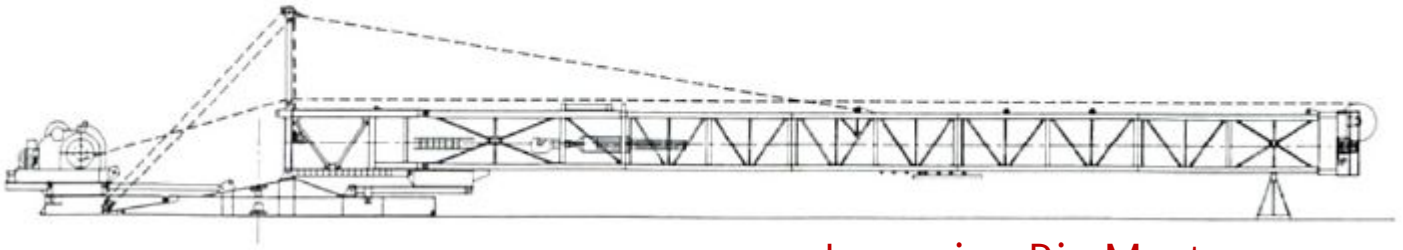
- Cementing Unit
- Air Drilling Unit
- Mud Logging Unit
- MWD Cabin

Well-Pad Facilities :

- Water & Mud Ponds
- Drainage System
- Temporary Cutting Bunker
- Access for Heavy Equipment
- Parking Area
- Muster Points

Site Construction and Rig Layout





Lowering Rig Mast

Behind:

- Space for Rig Engine.

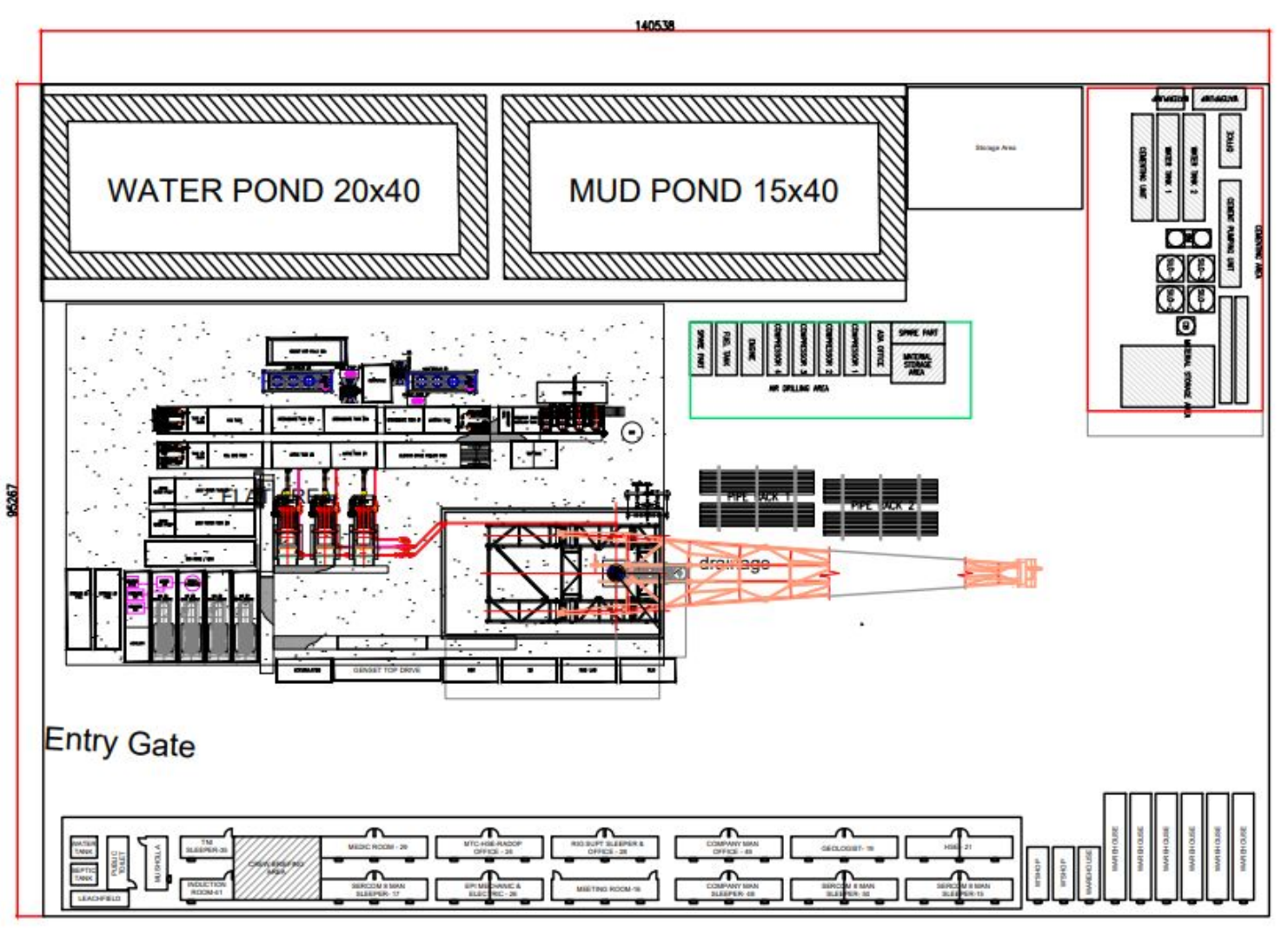
Front:

- Space for Lowering Rig Mast

Left and right side

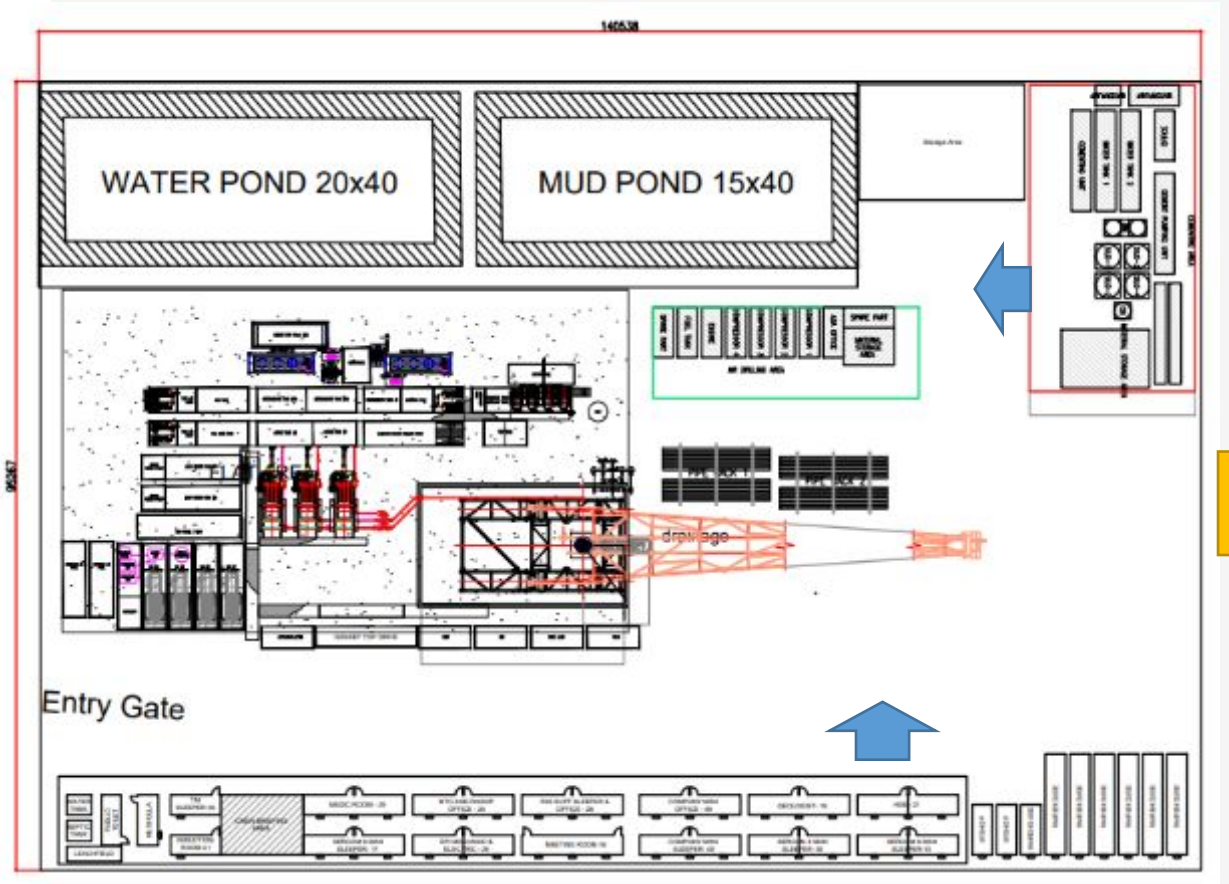
- Shale shakers and Mud Tanks
- Mini Camps
- Water Ponds
- Service Company
- Access for Heavy Equipment

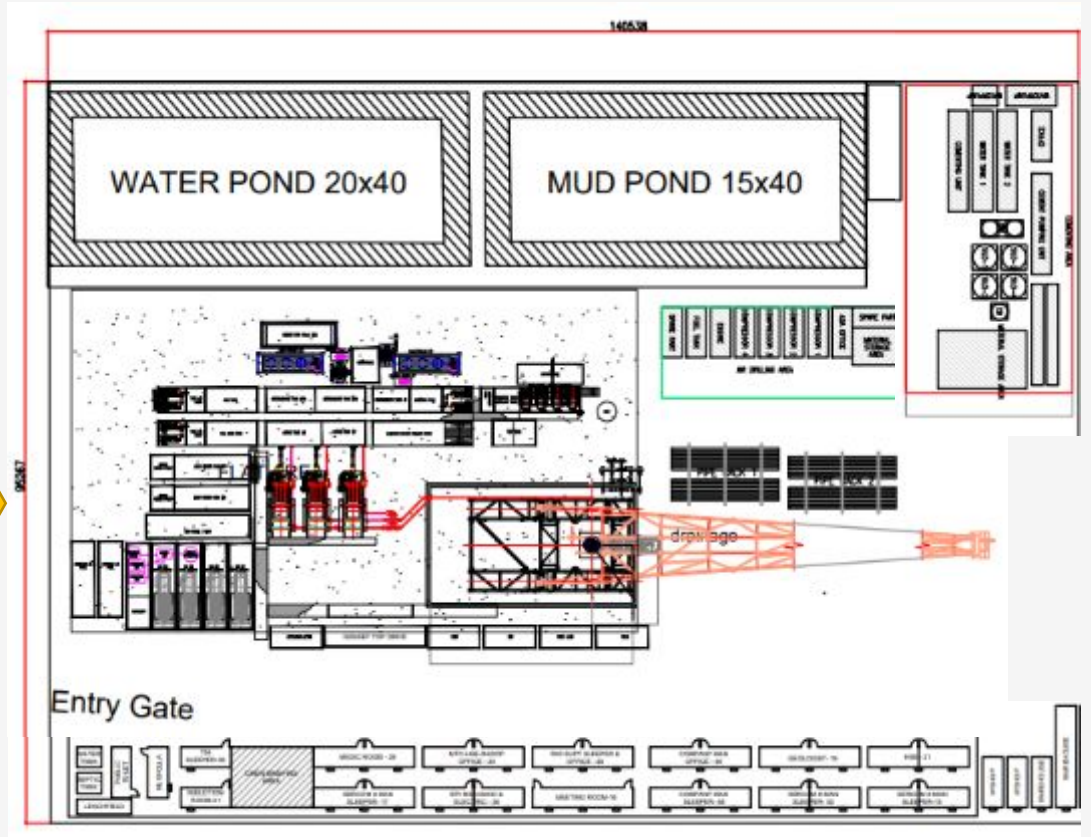
Pad-BB: Ideal Size WellPad



- Generic layout for new well pad
- Total size : 140 m x 95 m
- Interval between wells : 15 m (skidding)

Minimum Well Pad Size





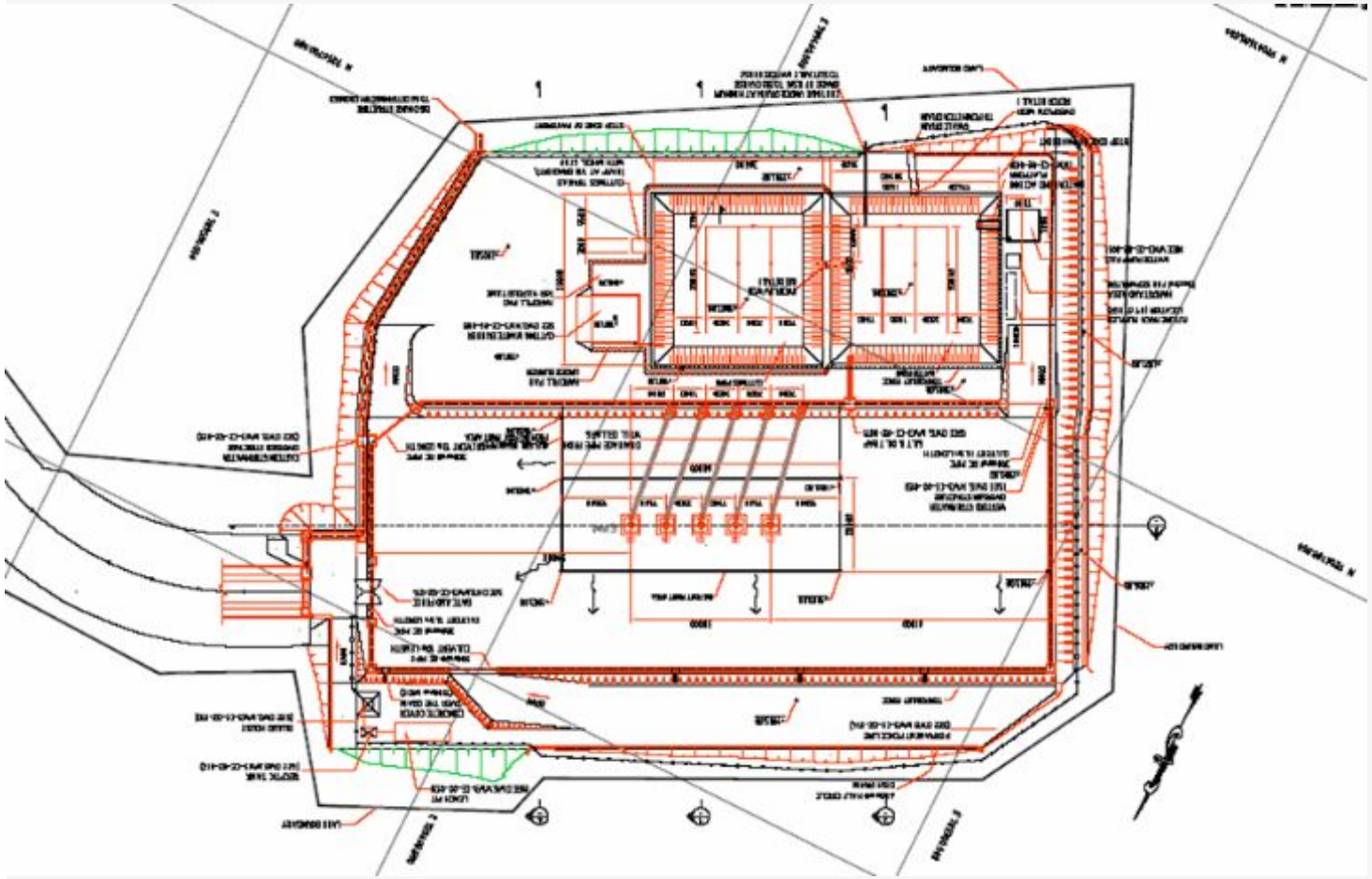
- Move Cementing Equipment
- Reduce access road
- Reduce numbers of Mini Camps
- Smaller water ponds
- No Water Pond

Site Construction – Geothermal Well



- Smaller well pad area due to in mountain condition.
- Surface hard rock
- Additional Space for water ponds: **5000 – 8000 m3**
- Safety Reason: Minimum number of Mini Camps, less rig personnel stay at rig, i.e H2S hazard
- Space for simultaneous operation with Well Test
  - Pipeline
  - Separator
- Rig with skidding system is a must.
- Ensure Rig has additional power cable adjust rig equipment lay out.

Optimum Well Pad Design



Typical Rig Site Layout

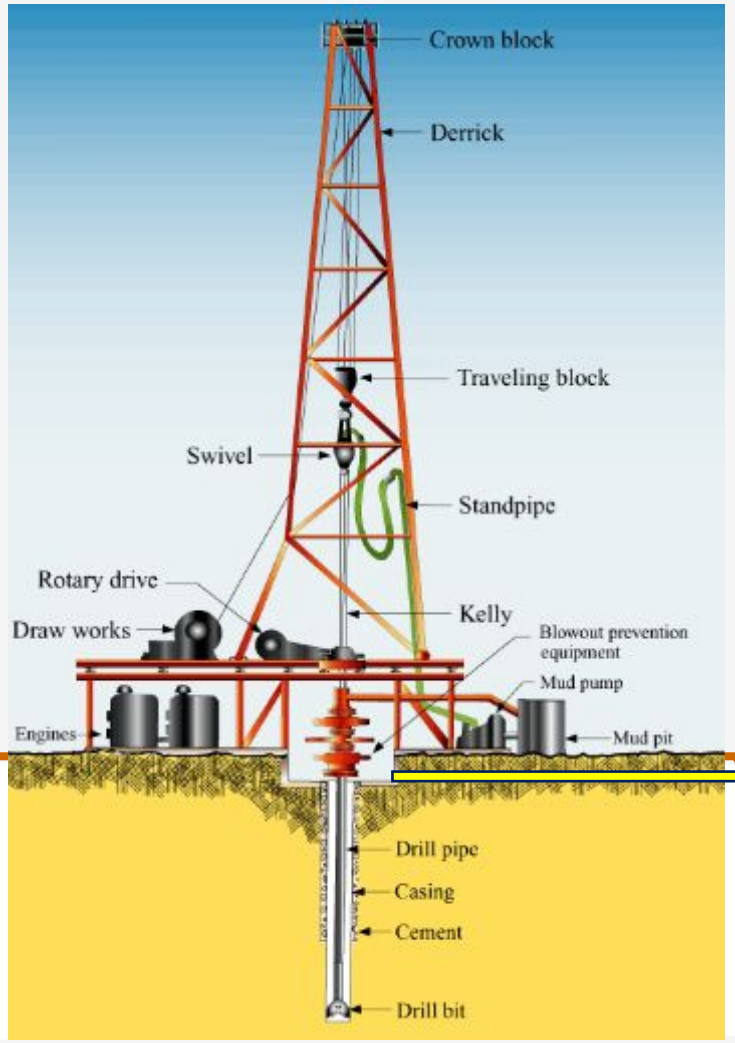
- 5000-8000 m3 water pond
- Lower elevation for cutting and water pond
- Pre design for well test pipe line layout
- Proper drainage and control box
- Direct Under ground drainage from Cellar to cutting Pond



Elevated Well-Pad Design







Drainage Line from Cellar to Mud Pond

Settling Pond

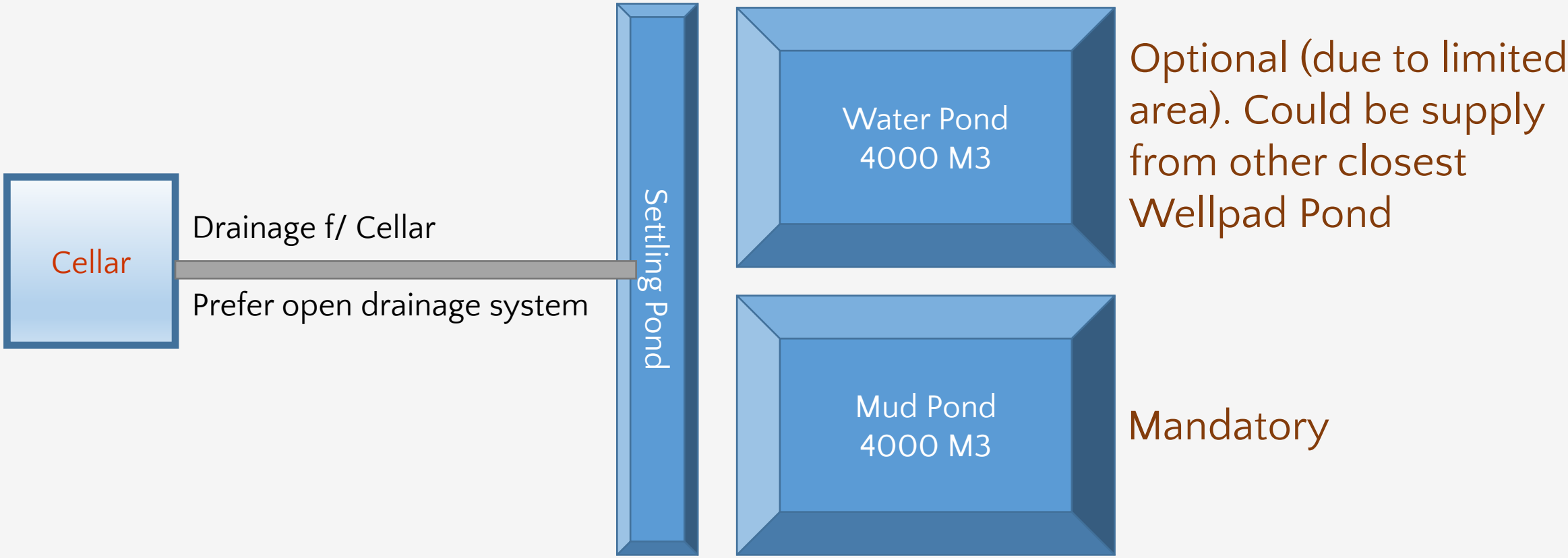
Water and Mud Pond

Layout higher elevation for Rig Mini Camps

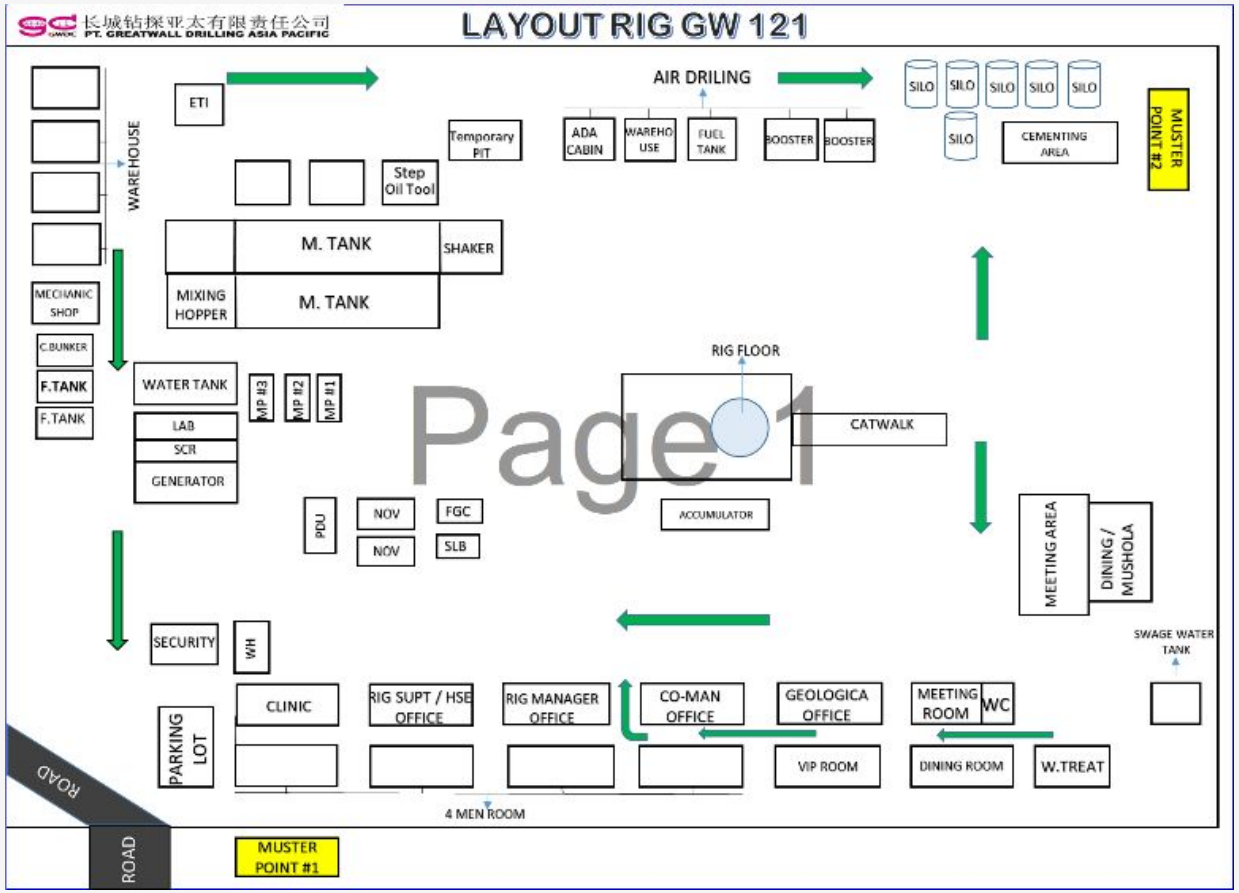




Mud Pond Size and Design

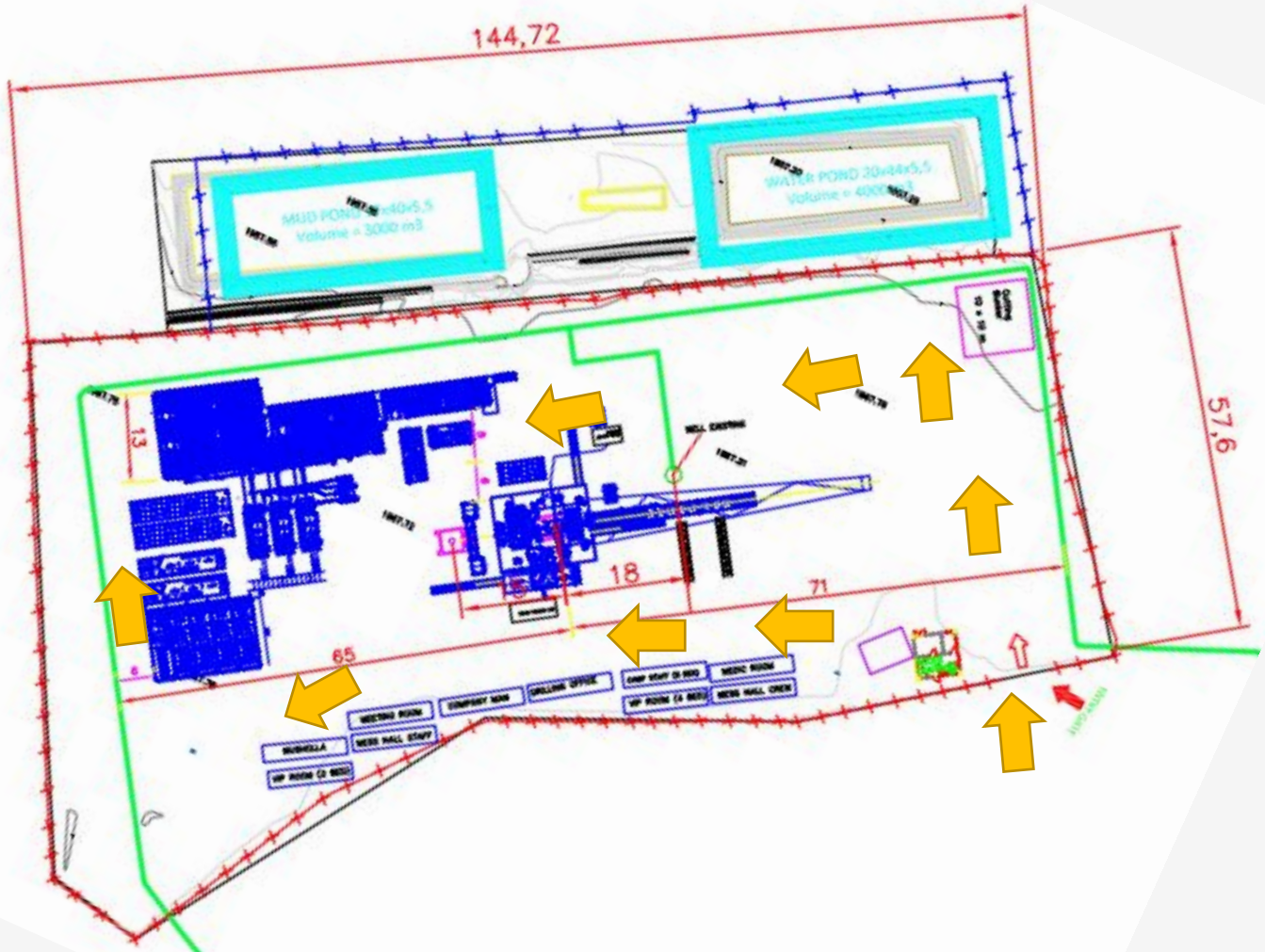


Rig Layout: Evacuation Route

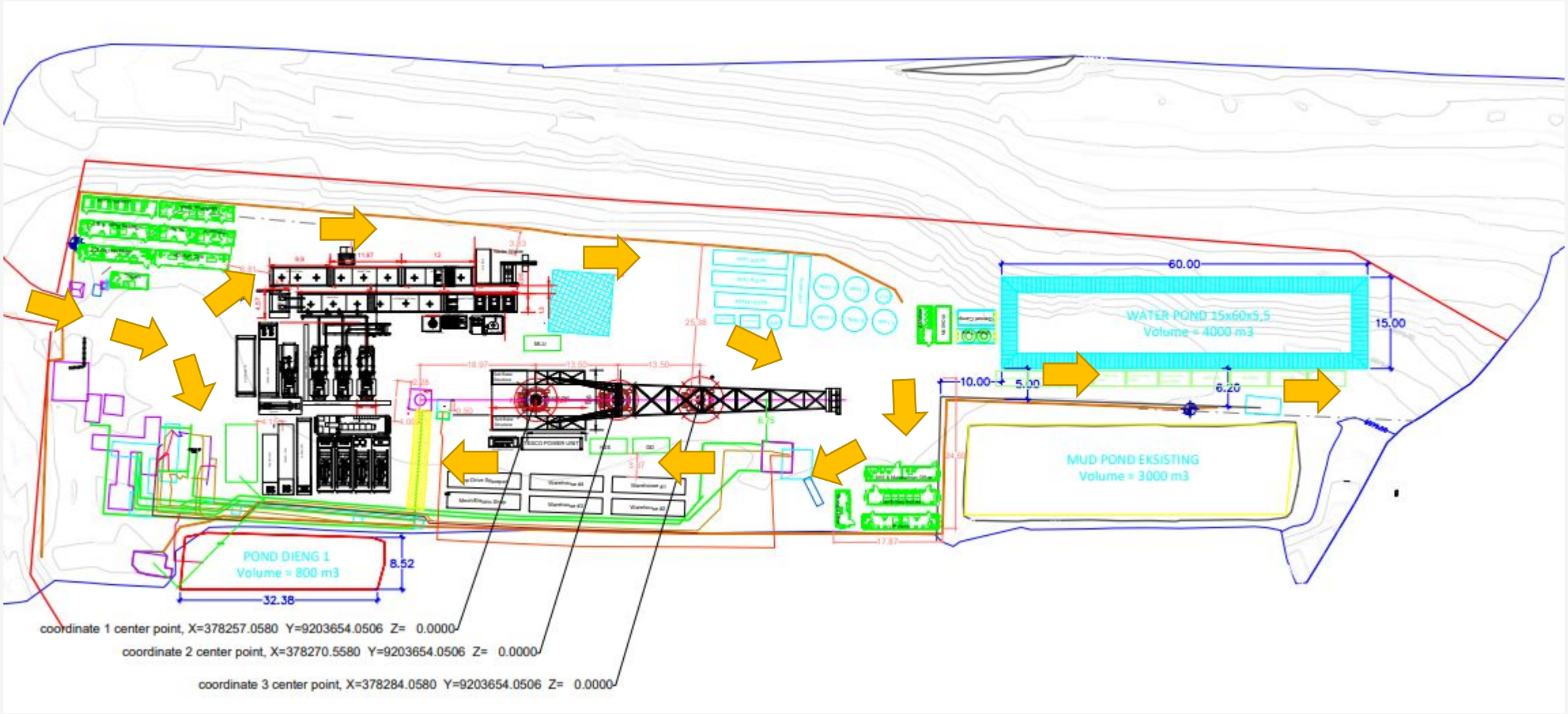




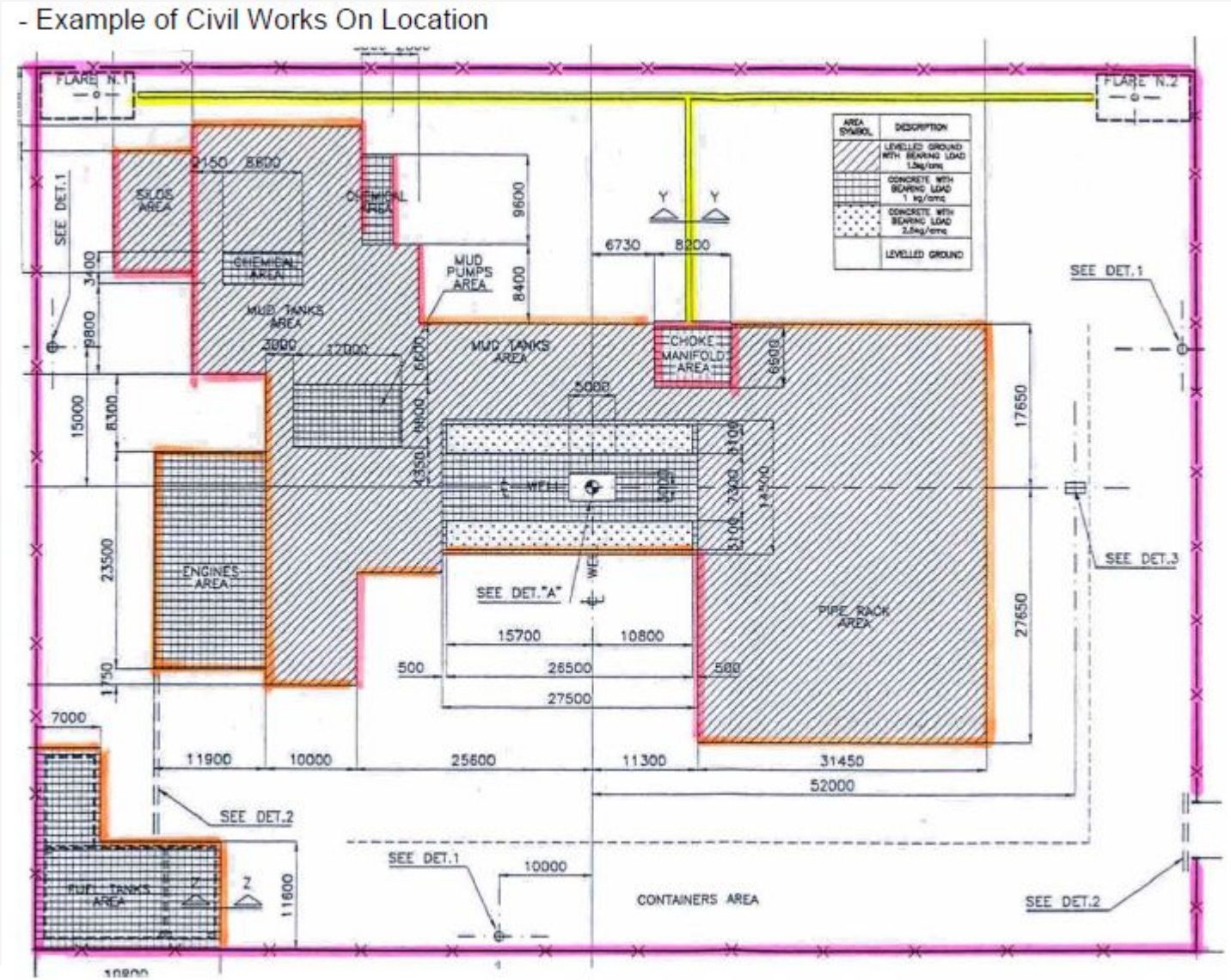
Rig Layout: Heavy Equipment Route



Rig Layout: Heavy Equipment Route

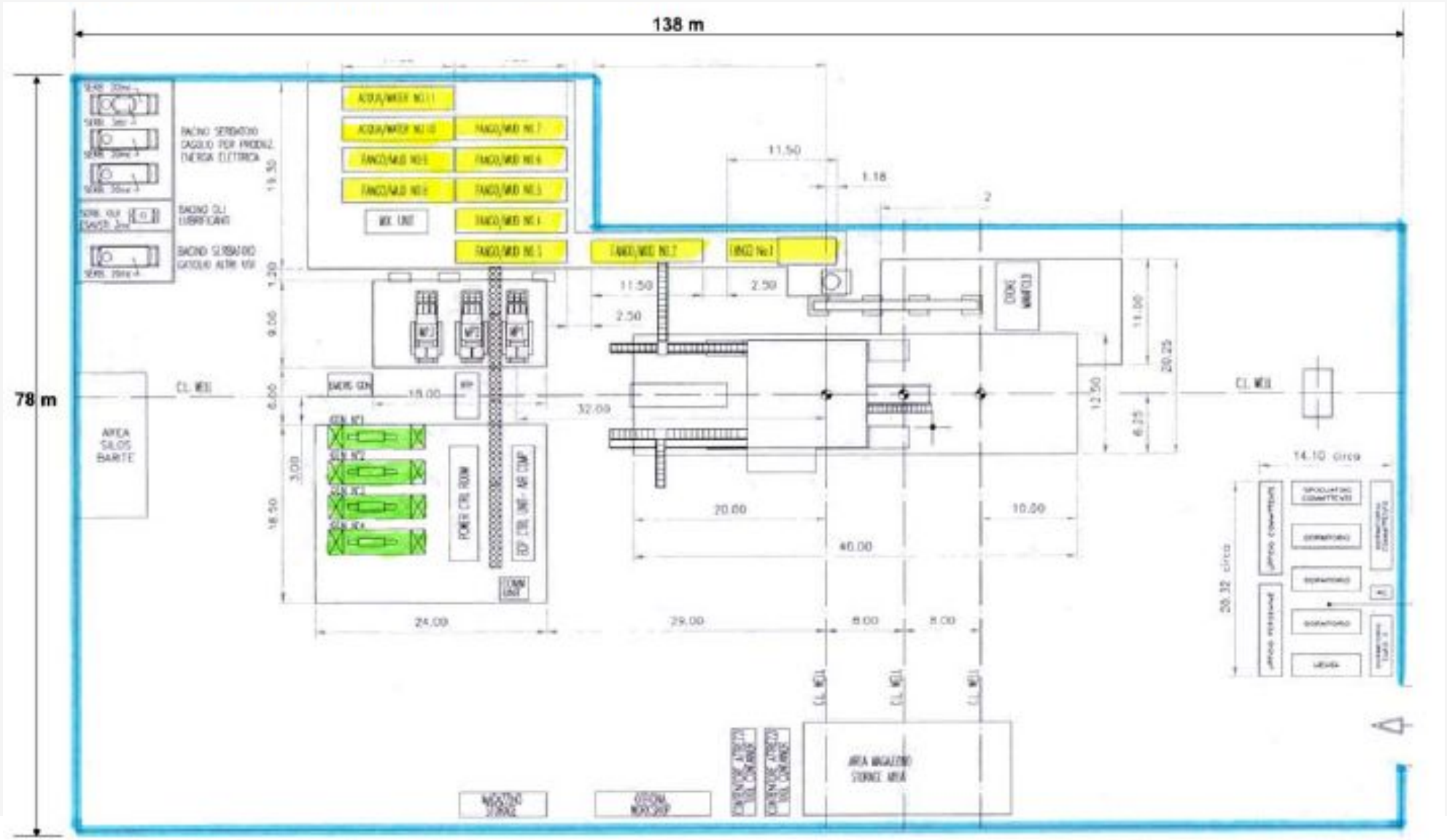


Example of Civil Works On Location

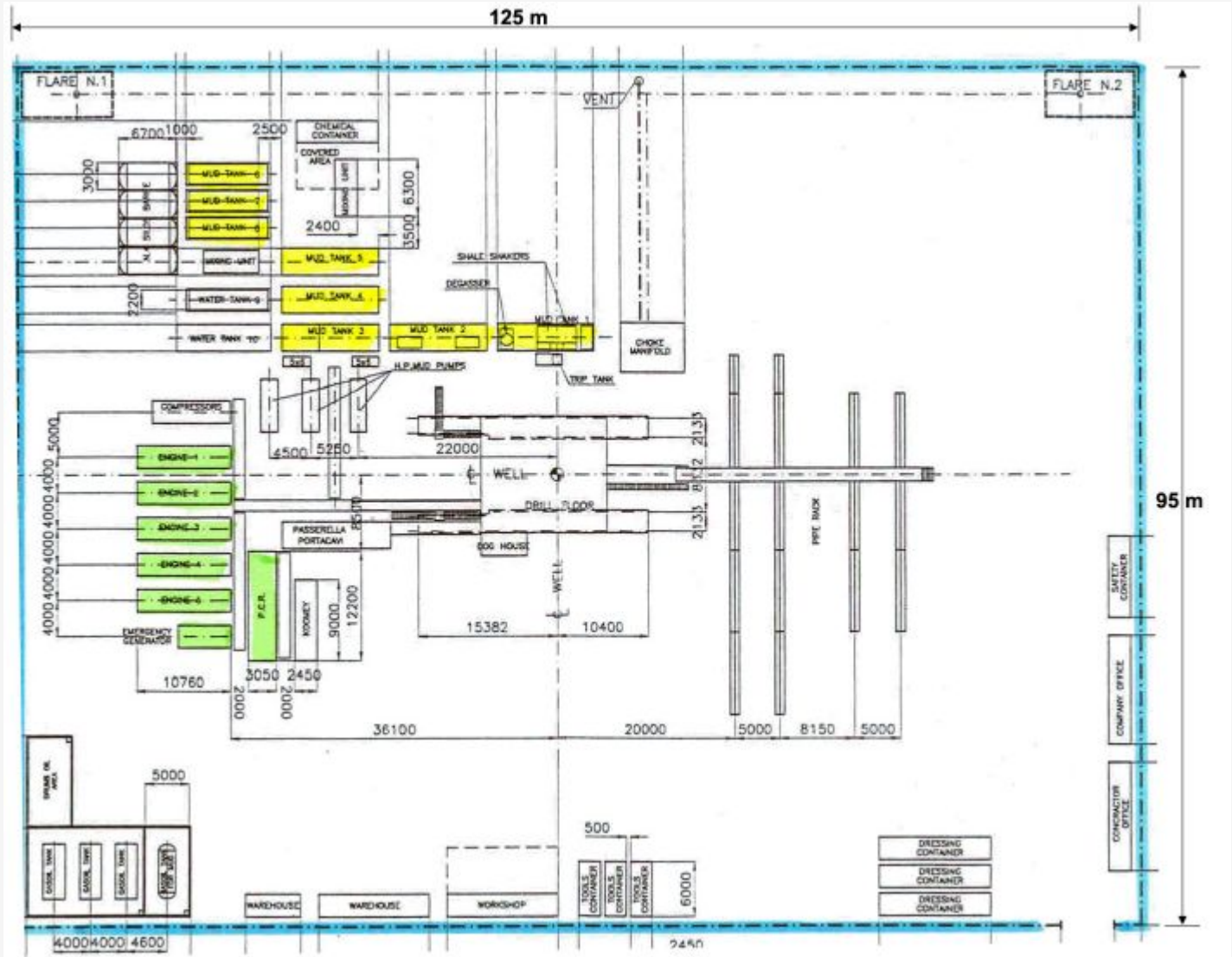


- Soil Bearing Load for Rig Foot Print.
- Area for leveling and compaction:
  - Mud tanks
  - Mud Pumps
  - Rig engine
  - Cementing
- Drainage system
- **Cutting bunker**
- **Cellar design**
- Water and cutting pond Capacity and elevation
- ~~Flare Pit area~~
- Mini camps area and septic tank
- Master Point Area
- Distance between well
- Etc……

Example of 3 Wells Cluster for 2000 HP Rig

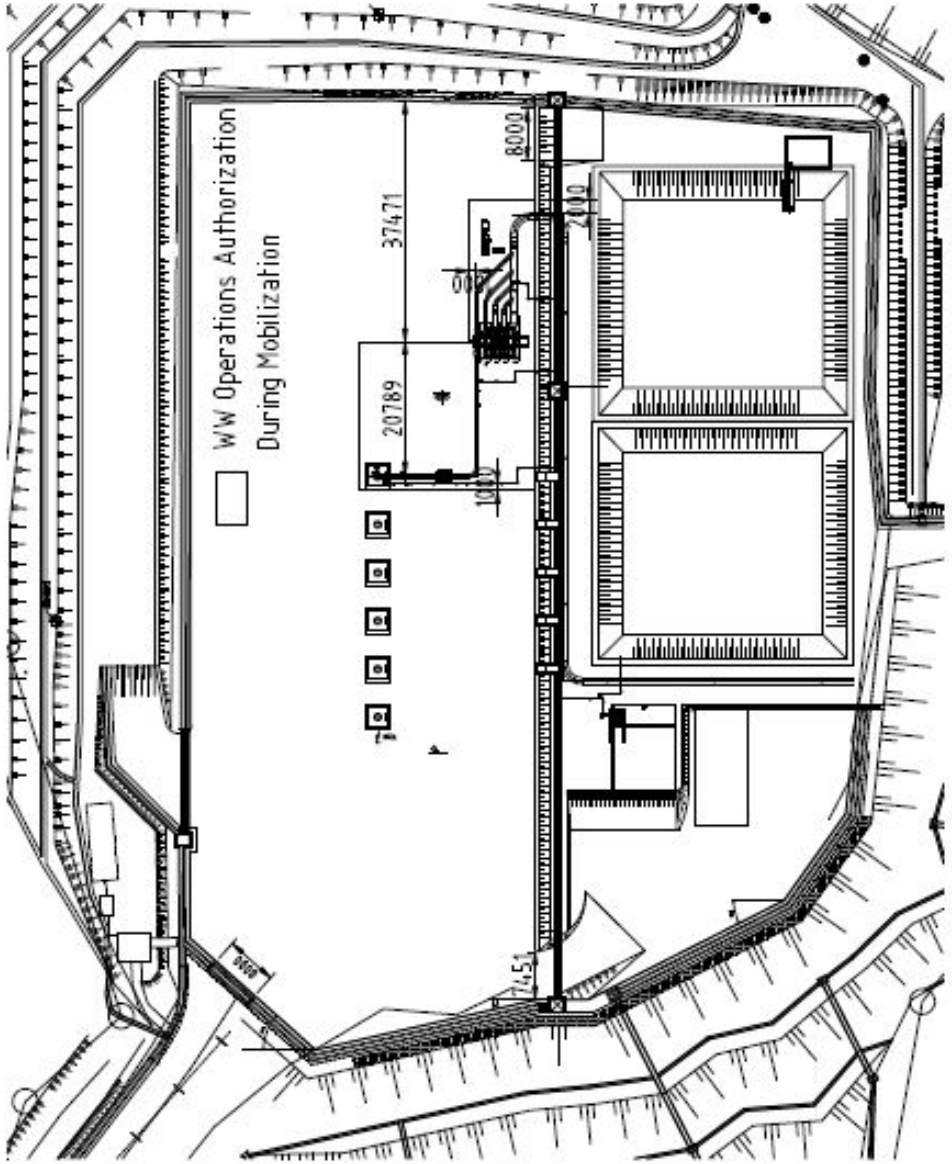


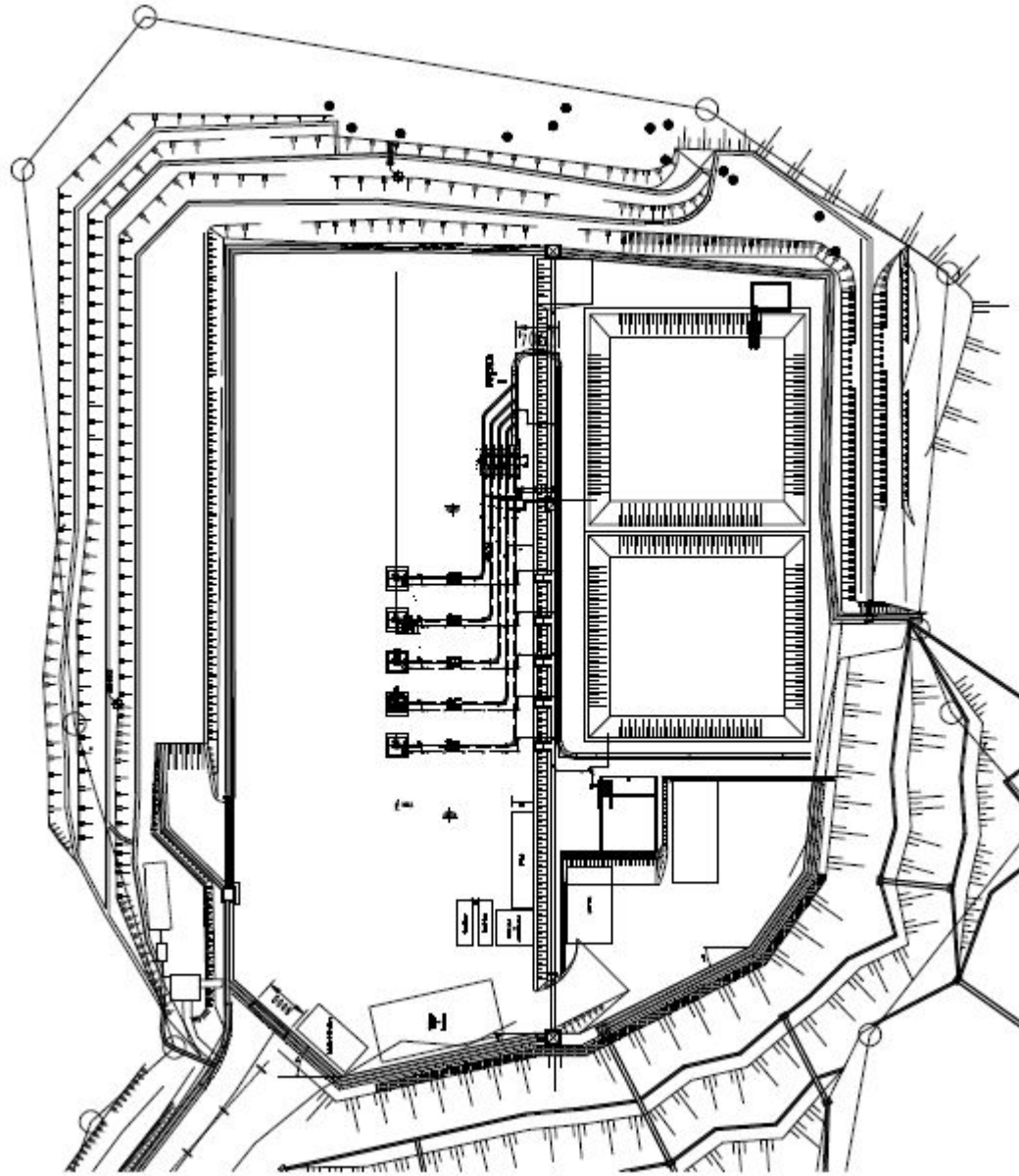
Example of single well for 3000 HP Rig



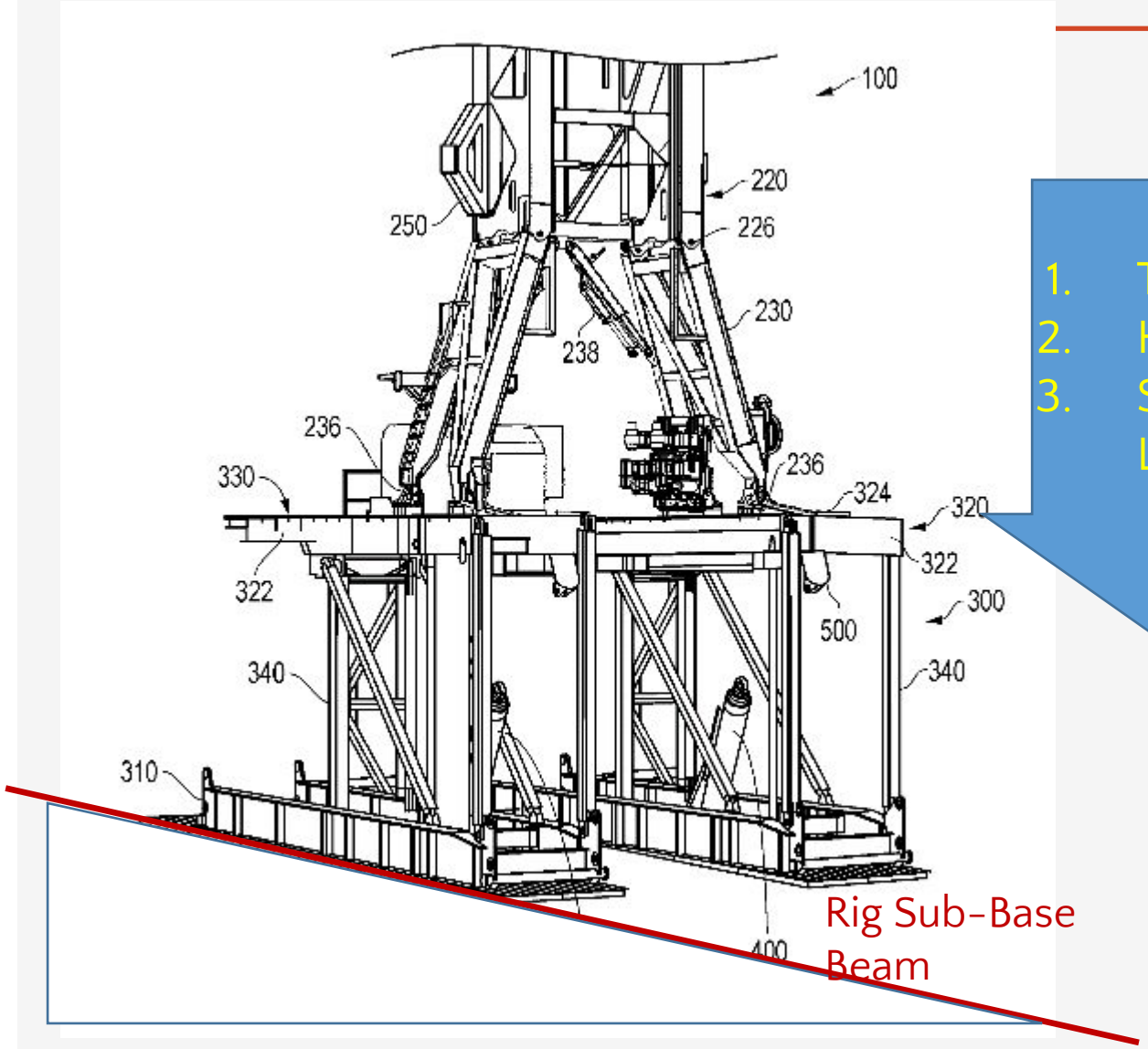
Ideal Geothermal Well Pad – Star Energy

- Mini camp area could be higher elevation than rig elevation.
- Distance between Cellar □ Well accessibility after being drilled
- Water flow by gravity to water pond.
- SIMOP: Pipeline layout not interfere with Rig Equipment layout.





Soil Bearing Load for Rig Sub-base



1. Total Weight
2. Hook Loads
3. Set Back Load

- Rig Foot-print elevation tolerance
- Soil Bearing Load Requirement

D2P2 Well Pad Minimum Soil Bearing Pressure Requirement for Rig Foot Print :

- 4500 PSF

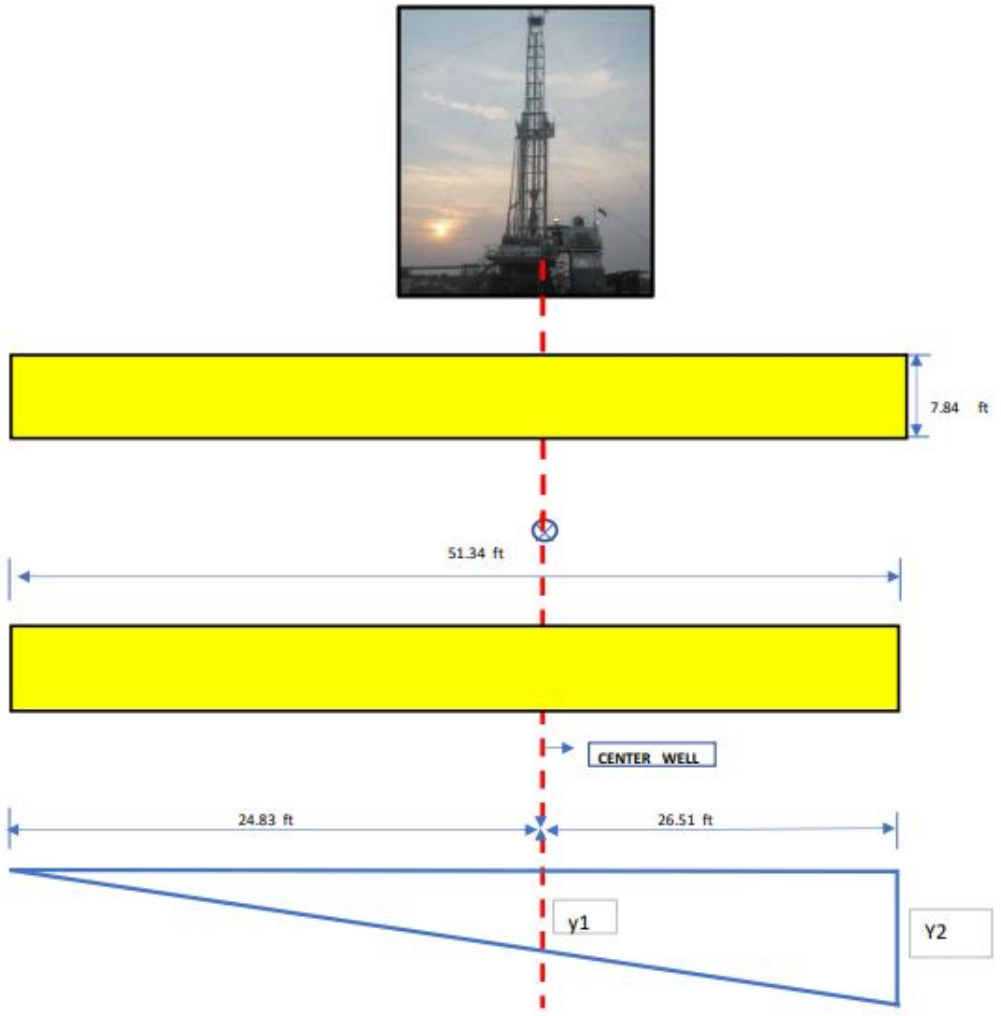
Rig Foot Print Soil Bearing Load

GROUND BEARING PRESSURE CALCULATION (PRA-09)

INPUT DATA'S

NO	DESCRIPTION	WEIGHT (Ton)
1	CROWN BLOCK W/MAST SECTION	9.0
2	RIG MAST UPPER	9.0
3	RIG MAST MIDDLE	9.0
4	RIG MAST LOWER	9.0
5	RIG MAST BOTTOM DS	17.5
6	RIG MAST BOTTOM ODS	17.5
7	"A' LEG SECTION	9.0
8	SUB BASE SUBSTRUCTURE + SKID DRAWWORK	12.0
9	SUB STRUCTURE DS	12.0
10	SUB STRUCTURE ODS	12.0
11	MONKEY BOARD	2.5
12	CROSS BEAM "A" LEG	1.0
13	CROSS BEAM FRONT FLOOR	10.0
14	DRAWWORK BEAM	1.0
15	FLOOR PLATE	1.0
16	FLOOR STAIRS,STAND PIPE	5.0
17	DOG HOUSE BUILDING	7.0
18	DRAWWORK GARDNER DENVER DW1100	30.0
19	2 UNIT DC MOTOR GE- 752 F/DRAWWORK	3.0
20	TRAVELING BLOCK G/National 660-G-500	9.2
21	ROTARY TABLE AND COMPONENTS	10.0
22	"V" DOOR	3.0
23	TOP DRIVE AND ITS TRACK / RAIL	9.0
24	SKIDING BEAM DS (6 set)	19
25	SKIDING BEAM ODS (6 set)	19
26		
	TOTAL WEIGHT (Ton)	246
	WEIGHT MAST (Lbs)	541,675
	ANOTHER DATA	
	HOOK LOAD (Lbs) 12 LINES	769,000
	SET BACK LOAD (Lbs)	500,000

DISTRIBUTION DATA RESULTS	
A. FROM CENTRE WELL TO RIGHT SIDE, FT	26.51
B. FROM CENTRE WELL TO LEFT SIDE, FT	24.83
E. SAFETY FACTOR FOR SUB BASE AREA	60%



PERHITUNGAN / CALCULATION

1 TOTAL LOAD DESIGN	=	1,810,675 lbs	3 BEARING PRESS. (Y1)	=	2,248.75 psf
2 SUB BASE AREA	=	805.19 ft2	4 BEARING PRESS (Y2)	=	4,649.64 psf

SF	45%	50%	55%	60%	65%	70%	75%	80%
Y1 (psf)	3260.69	3373.13	3485.56	3598.00	3710.44	3822.88	3935.31	4047.75
Y2 (psf)	6741.97	6974.46	7206.94	7439.42	7671.90	7904.38	8136.86	8369.35

Rig Footprint Soil Bearing Load

APEXINDO

LOAD BEARING CALCULATION
RIG #5

No	Description		Ton			Pounds (use 9.2 ppg)	Total Weight (Pounds)
1	Mast		40			88185	
2	Substructure		140			308647	
3	Drawwork and Support Rotary Table		70			154323	
4	V-Door TDS		13.5			29762	
5	Travelling Block Air Hoist		20			44092	
6	Catwalk Dog House		10			22046	
7	Setback and casing load cap					1300000	
8	Hook Load Cap.					1125000	
9	Skidding Beam		40			88185	
10	Matting Boards 20'x8' @2000 kg per each		12			26455	
11	Monkey Board		6			13228	3,199,924

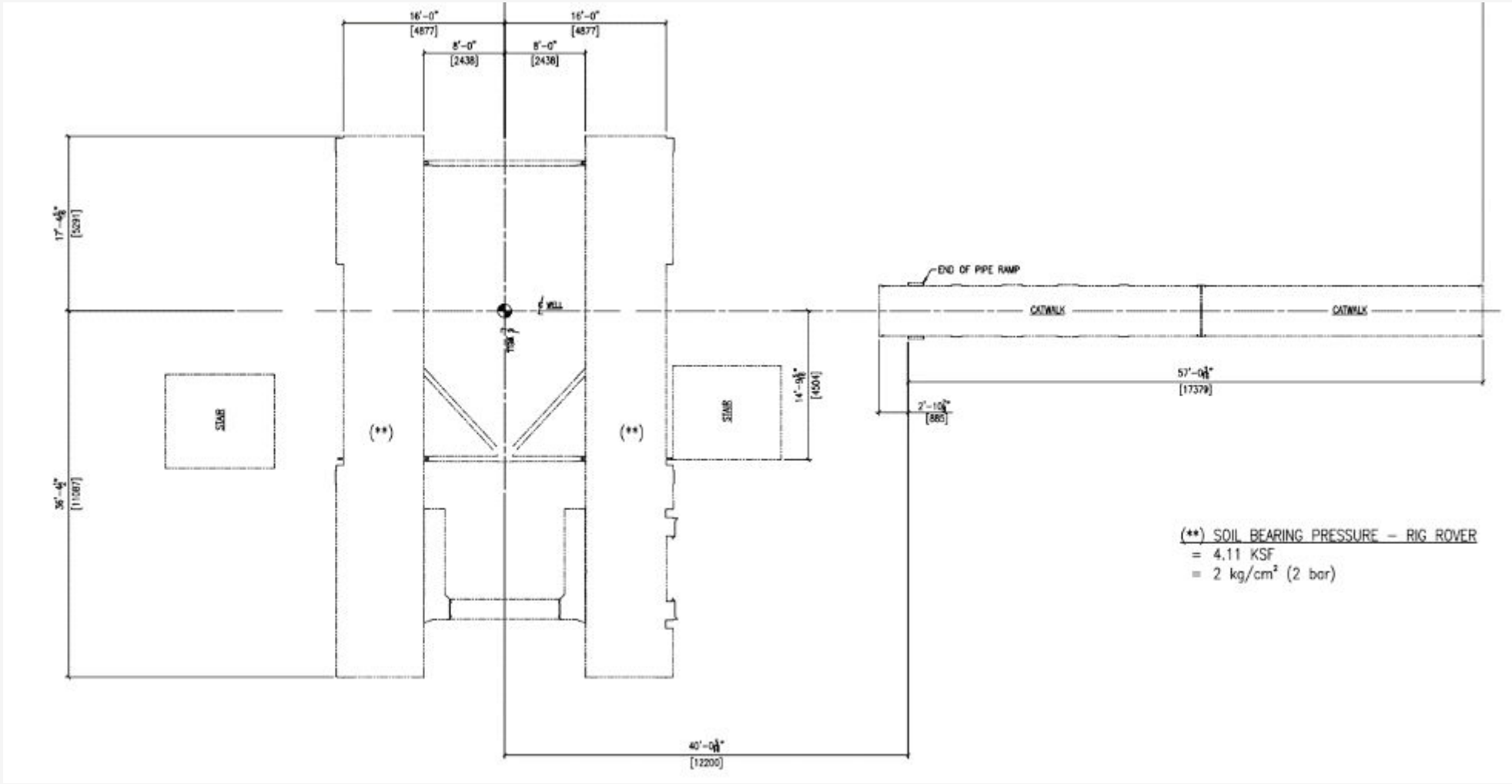
			feet	inch	Area (inch2)	Area (m2)	Area(ft2)
	Each Area Sub Base	L	50.125	601.5	21654	14	151
		W	3	36			
	Each Area Matting Board	L	60	720	146880	95	1023
		W	17	204			

Note:

		ft		
1 matting board	P	20		ft2
	L	8.5	Total area of 12 ea Matting Board is	2040
Total Matting Boars	ea	12		

Total Load Bearing/ Area with Matting = 1569 Lbs/ft2

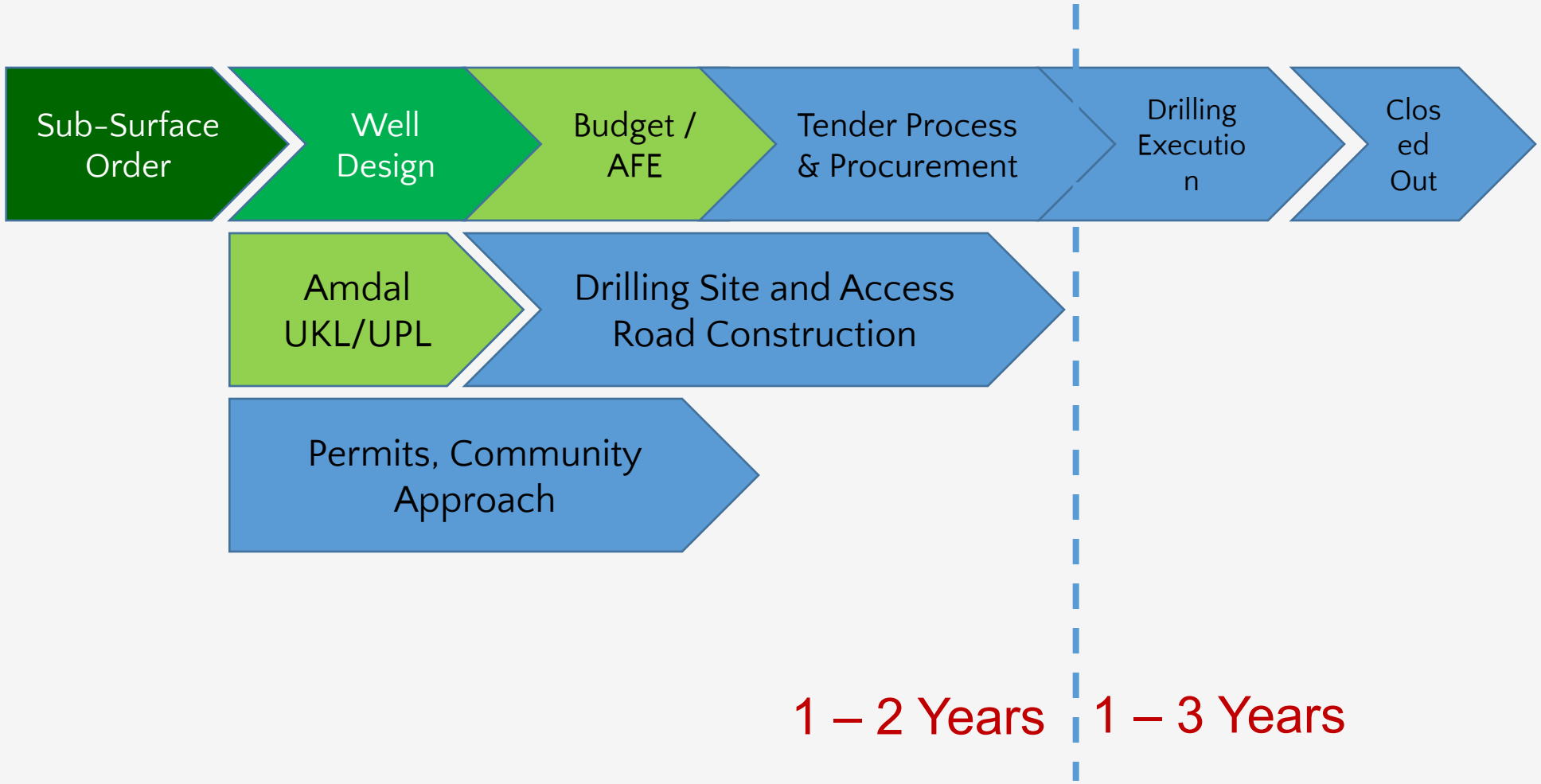
Contact Footprint For 147’ HMRIII Mast Model 27604



4.11 KSF

When we construct well-pad?

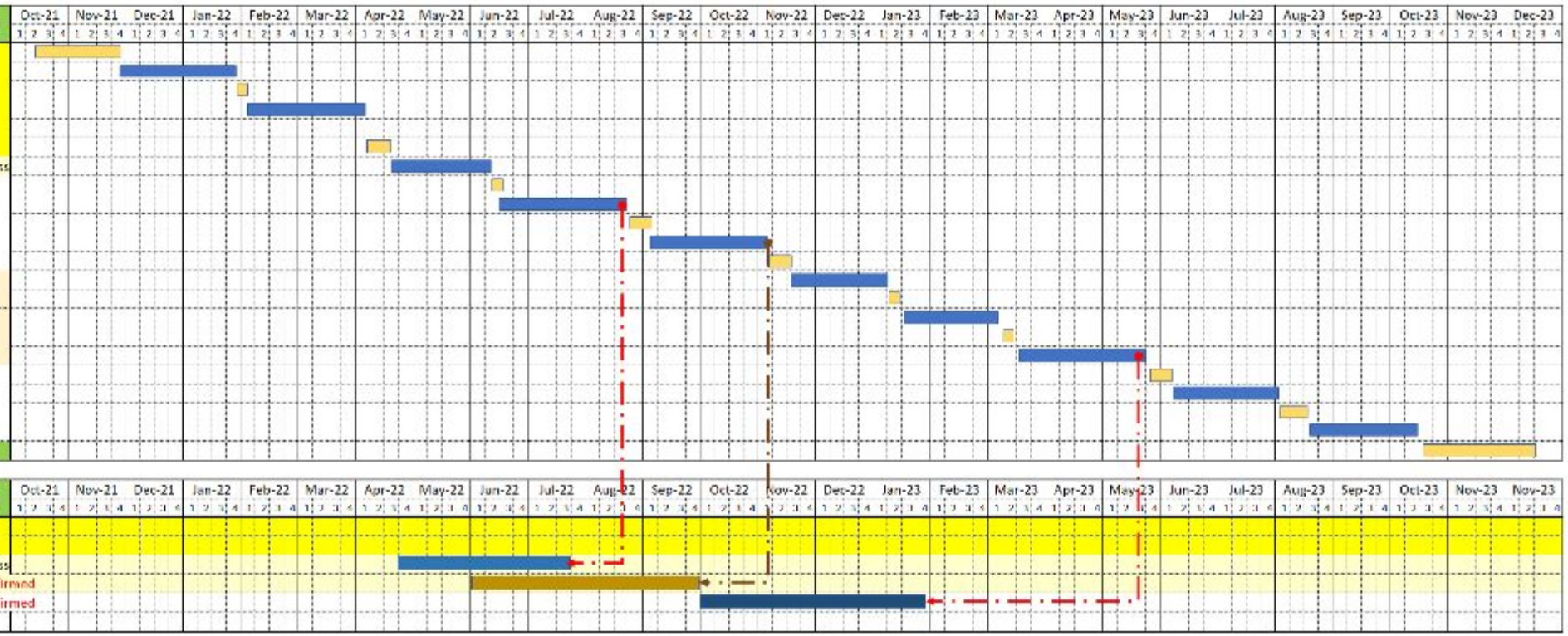
DRILLING PROJECT FLOW CHART



Drilling Schedule vs Well-Pad Construction

No	Drilling Activity - Dieng	Days	TD, m-MD	Start	End	Remark
1	Rig Mobilization & RU	43		12-Oct-21	24-Nov-21	Done
2	SLR-Q-31A	65	2600	24-Nov-21	28-Jan-22	
3	Skid and RU	4		28-Jan-22	1-Feb-22	
4	SLR-Q-31B	56	2900	1-Feb-22	29-Mar-22	
5.a	Rig PMS and Stand By	11		29 Mar 22	9 Apr 22	
5.b	RD, RM, RU	14		9-Apr-22	23-Apr-22	
6	SLR-T-9C	58.5	2839	23-Apr-22	20-Jun-22	On Progress
7	Skid and RU	5		20-Jun-22	25-Jun-22	
8	SLR-T-9D	58.5	2650	25-Jun-22	23-Aug-22	
9	RD, RM, RU	14		23-Aug-22	6-Sep-22	
10	SLR-I-7D	58.5	3000	6-Sep-22	3-Nov-22	
11	RD, RM, RU	14		3-Nov-22	17-Nov-22	
12	SLR-29D	58.5	2700	17-Nov-22	15-Jan-23	
13	Skid and RU	5		15-Jan-23	20-Jan-23	
14	SLR-29B	58.5	2800	20-Jan-23	19-Mar-23	
15	Skid and RU	5		19-Mar-23	24-Mar-23	
16	SLR-29C	58.5	2800	24-Mar-23	22-May-23	
17	RD, RM, RU	14		22-May-23	5-Jun-23	
18	SLR-Q-30B	58.5	2500	5-Jun-23	2-Aug-23	
19	RD, RM, RU	14		2-Aug-23	16-Aug-23	
20	SLR-I-7E	58.5	2600	16-Aug-23	14-Oct-23	
21	RD and Demob the Rig	60		14-Oct-23	13-Dec-23	

No	Civil Work - Dieng	Days	Start	End	Remark
1	Kontruksi Pad-31				Done
2	Kontruksi Pad-09				
3	Kontruksi Pad-07	90	20-Apr-22	19-Jul-22	On Progress
4	Kontruksi Pad-29	120	1-Jun-22	29-Sep-22	To be confirmed
5	Kontruksi Pad-30	120	29-Sep-22	27-Jan-23	To be confirmed
	Estimated days for Civil Works				

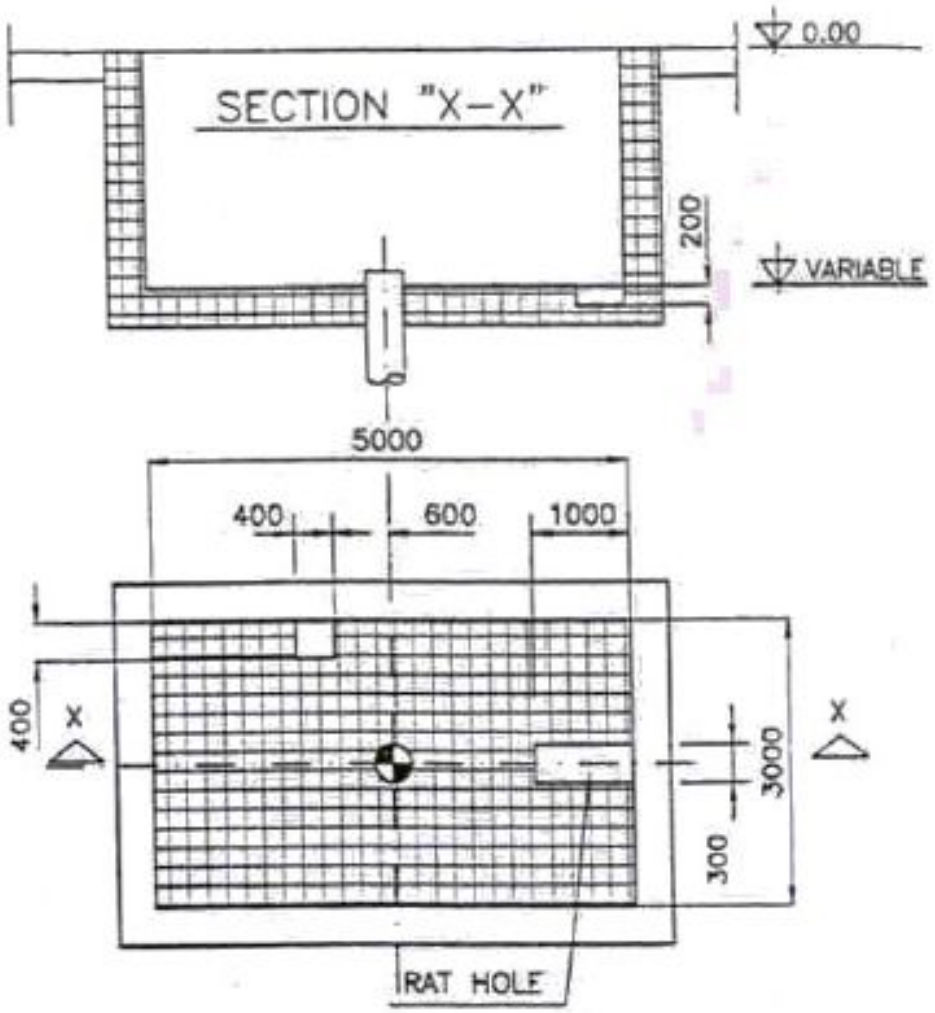


Cellar Design

Cellars Design

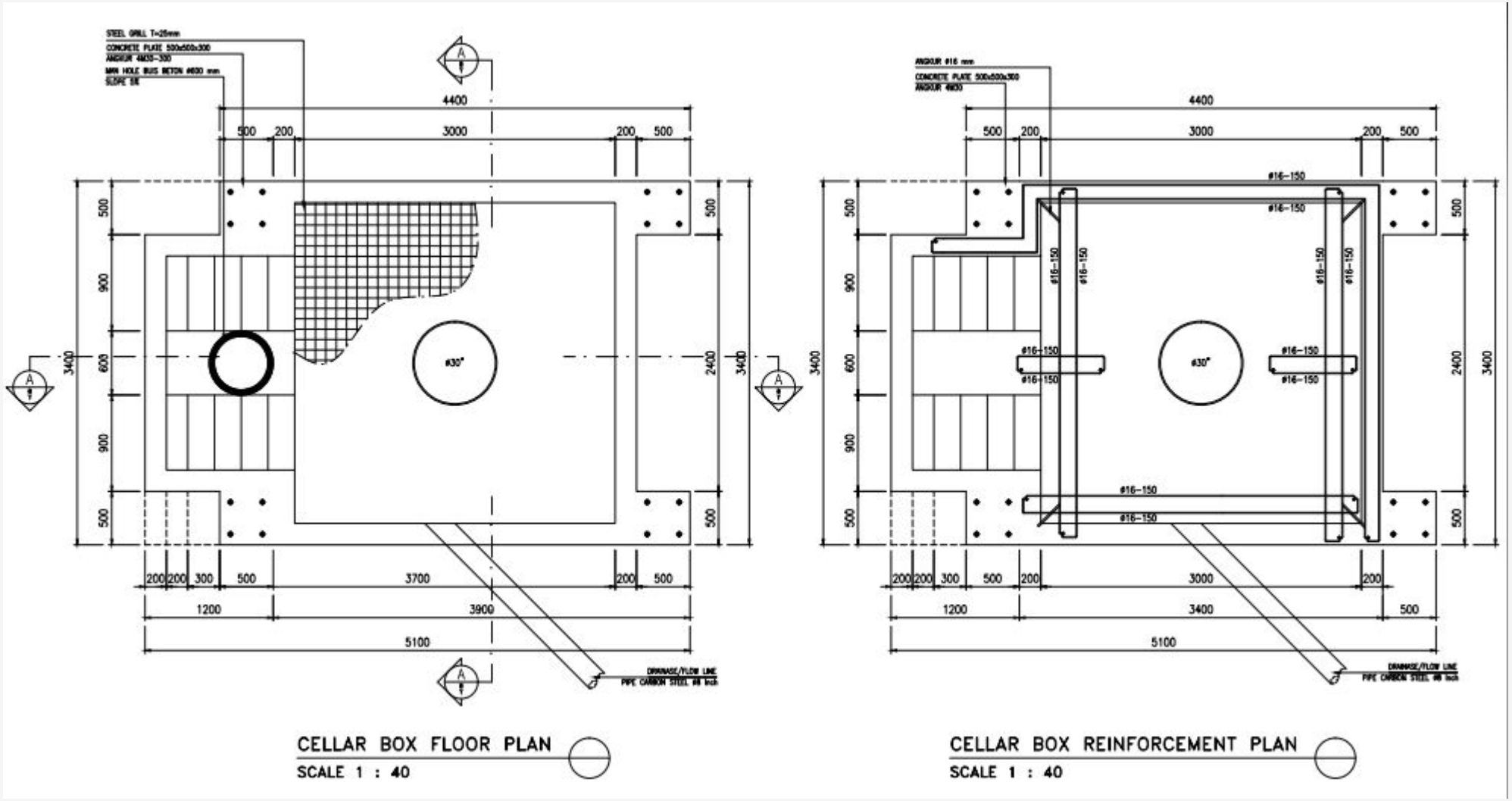
- Cellar breadth

Cellar breadth is usually decided with the Rig Contractor, considering well head, BOP and substructure. The cellar is usually cased in concrete to avoid collapse with the weight of the rig.

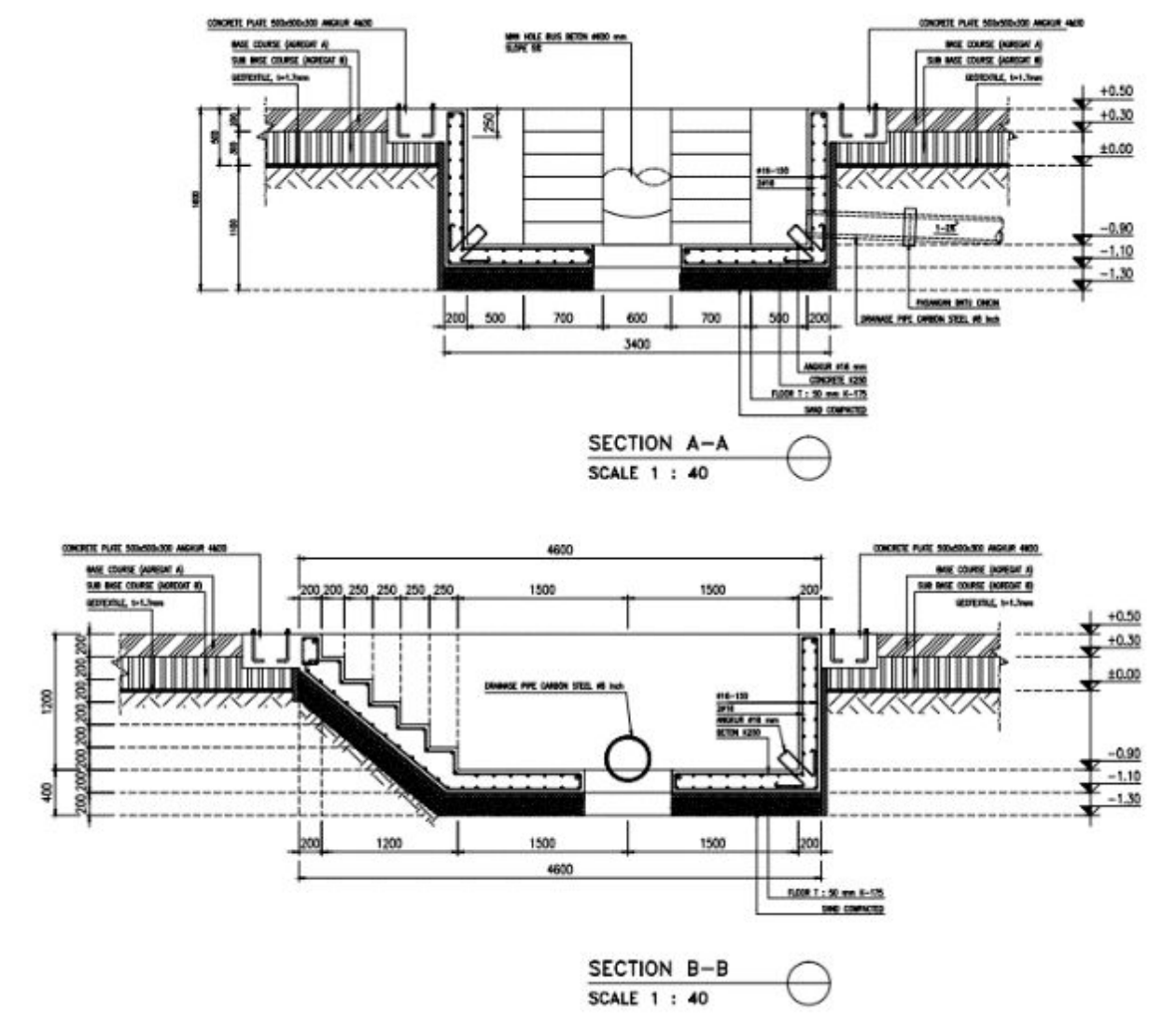


- Common size 3x3 m

Cellars Design-1: 3m X 3m X 1.2m depth



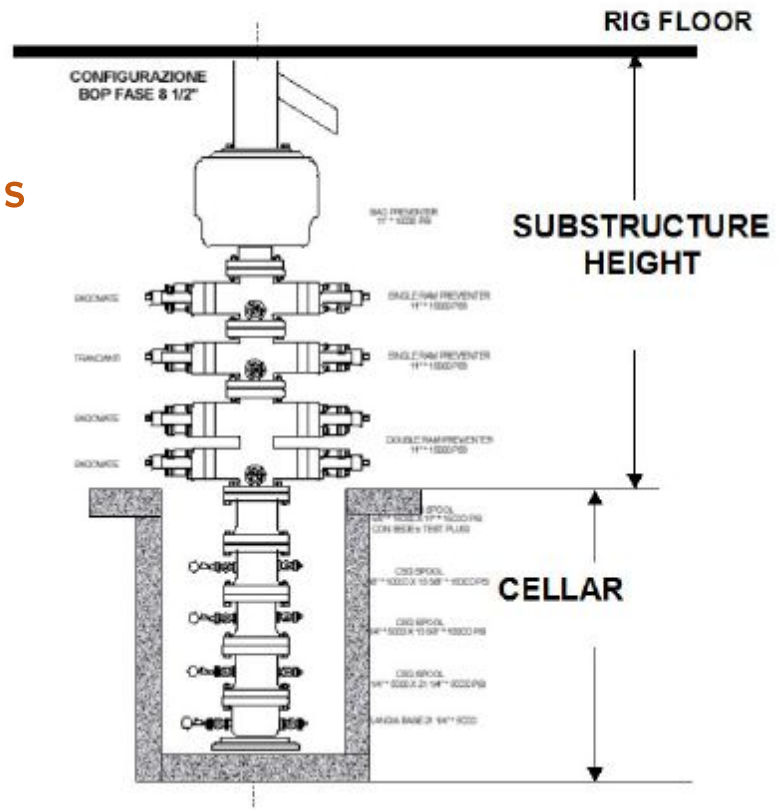
Cellars Design-1: 3m X 3m X 1.2m depth



Cellars Design

- Cellar depth

Cellar depth depends on substructure height, BOP and well head dimensions.

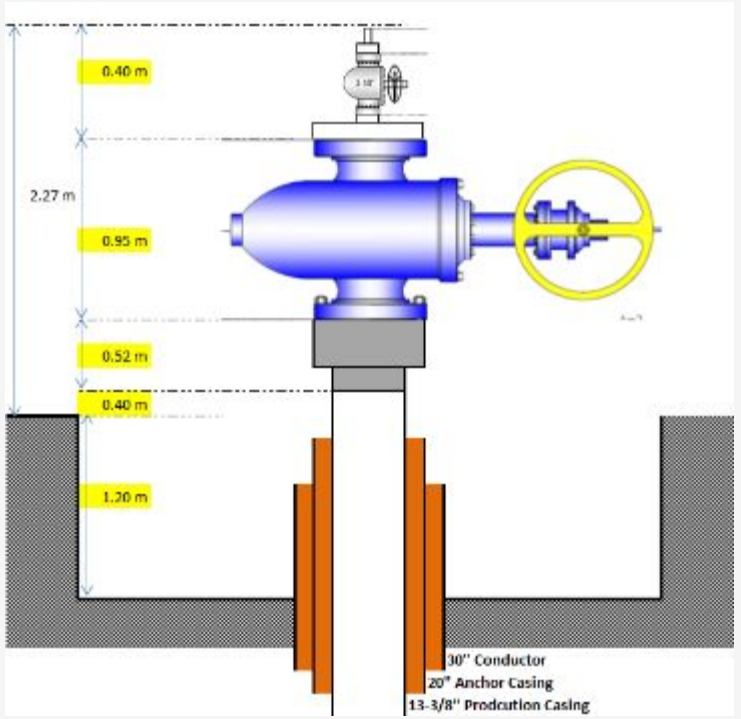


Common Practice for Oil/Gas Well:

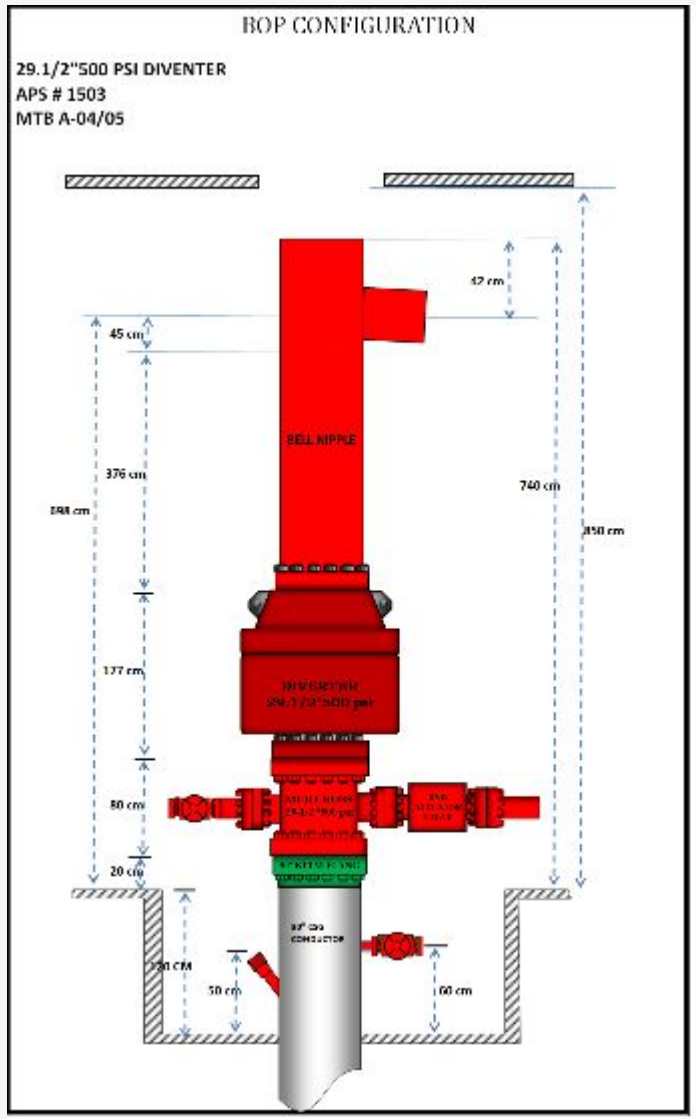
- Place wellhead inside cellar and X-Tree above ground elevation.

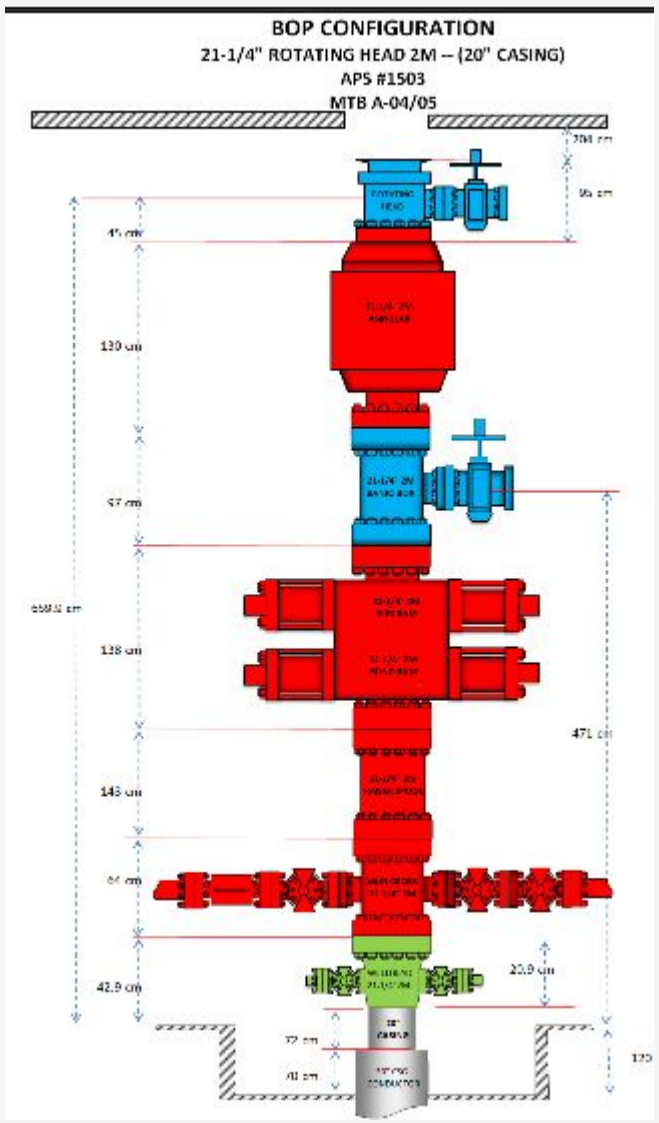
Common Practice for Geothermal Well:

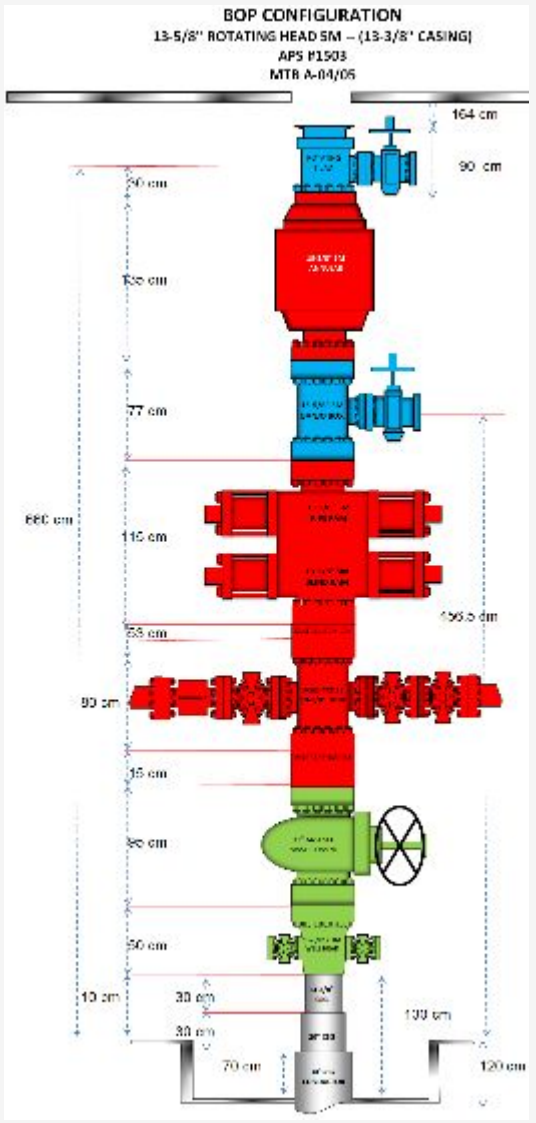
- Place wellhead above ground elevation.
- Easy to operated Master Valve (120 – 160 cm above GL)



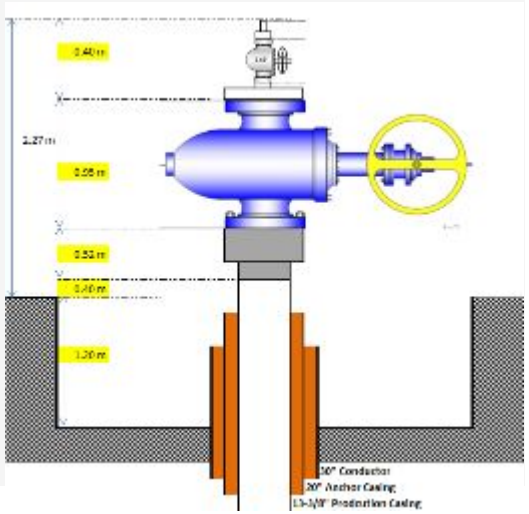
Cellars Design-1: 3m X 3m X 1.2m depth



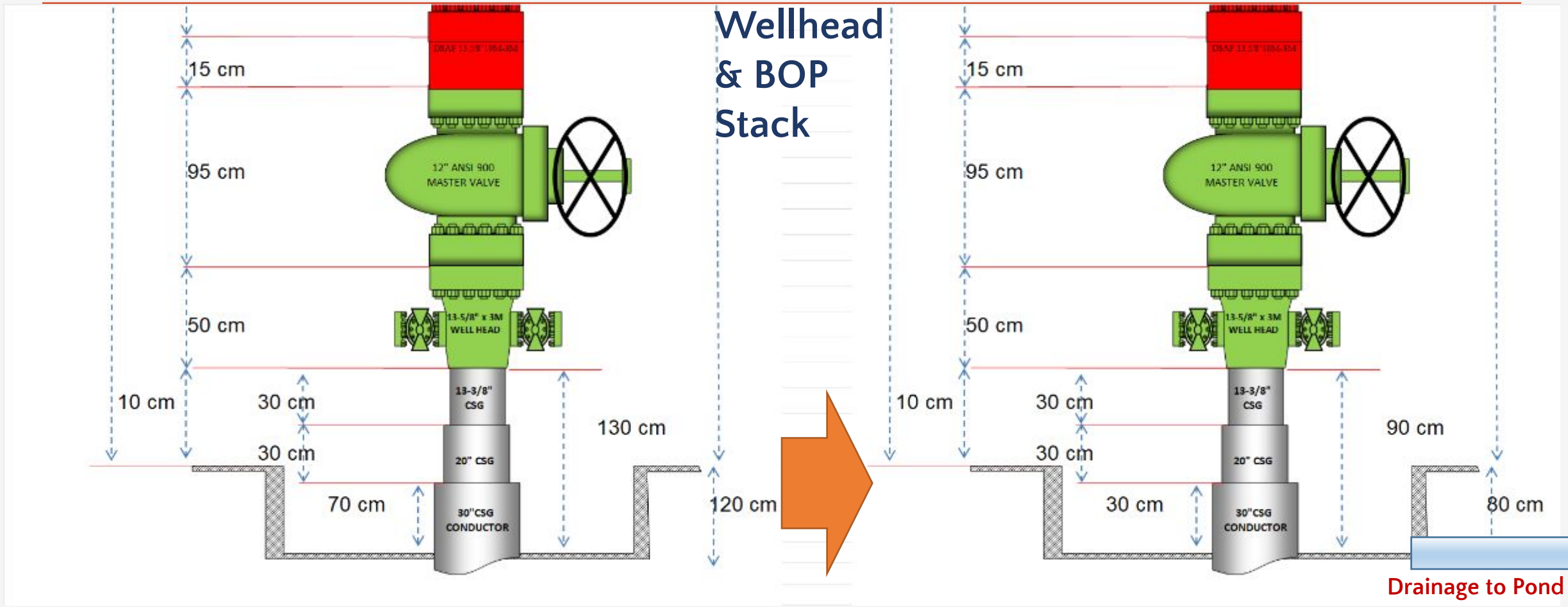




Wellhead & BOP Stack



Cellars Design-2: 3m X 3m X 0.8m depth



3m X 3m X 1.2m depth

3m X 3m X 0.8m depth

Cellars Design-2: 3m X 3m X 0.8m depth





Advantage:

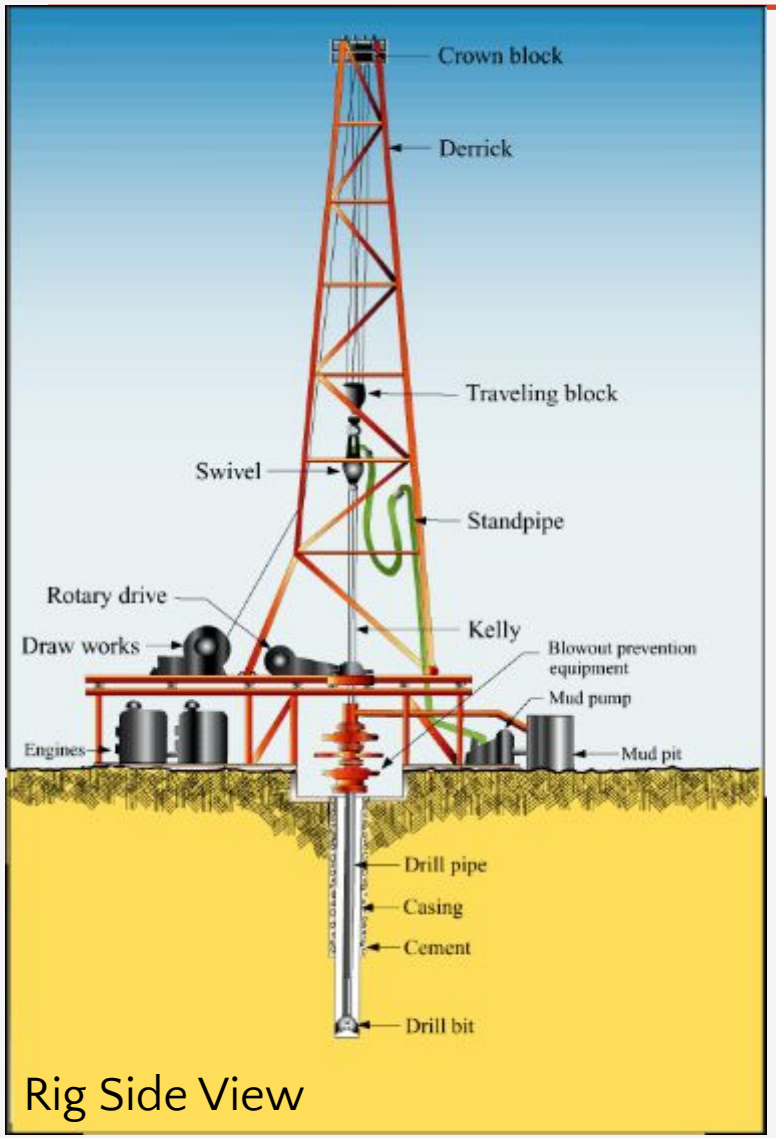
- More simple Less Construction Cost
- Always dry cellar
- No risk accumulated H2S

Integrated Cellars with Concrete For Rig Foot Print





Rig Picture

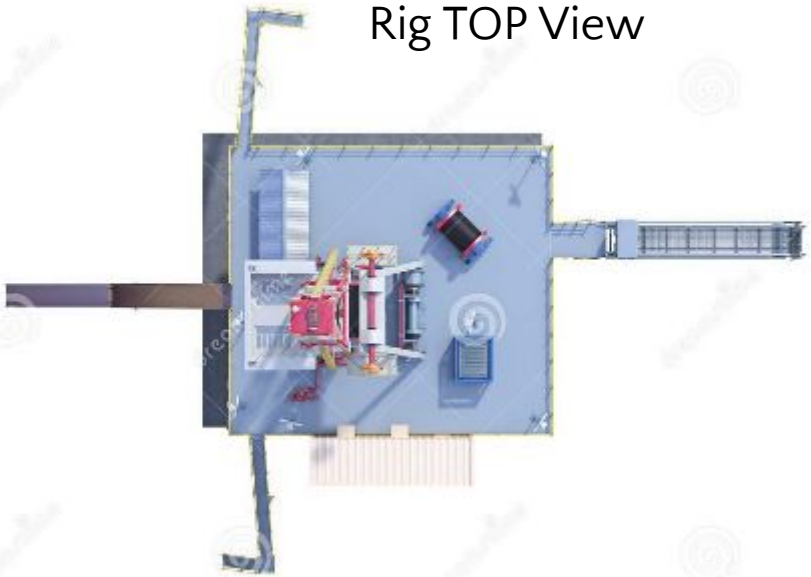


Rig Side View

Rig Side View



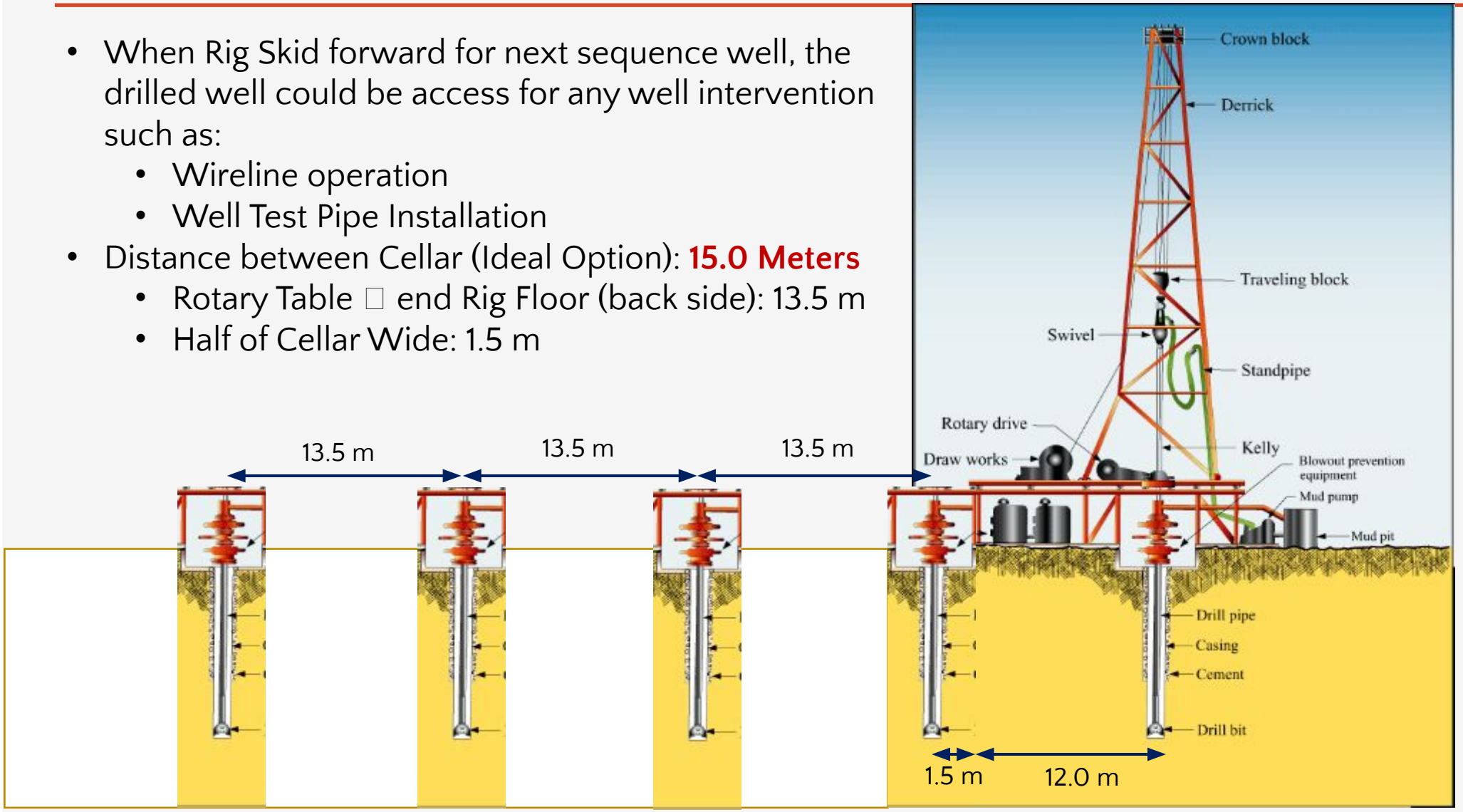
Rig TOP View



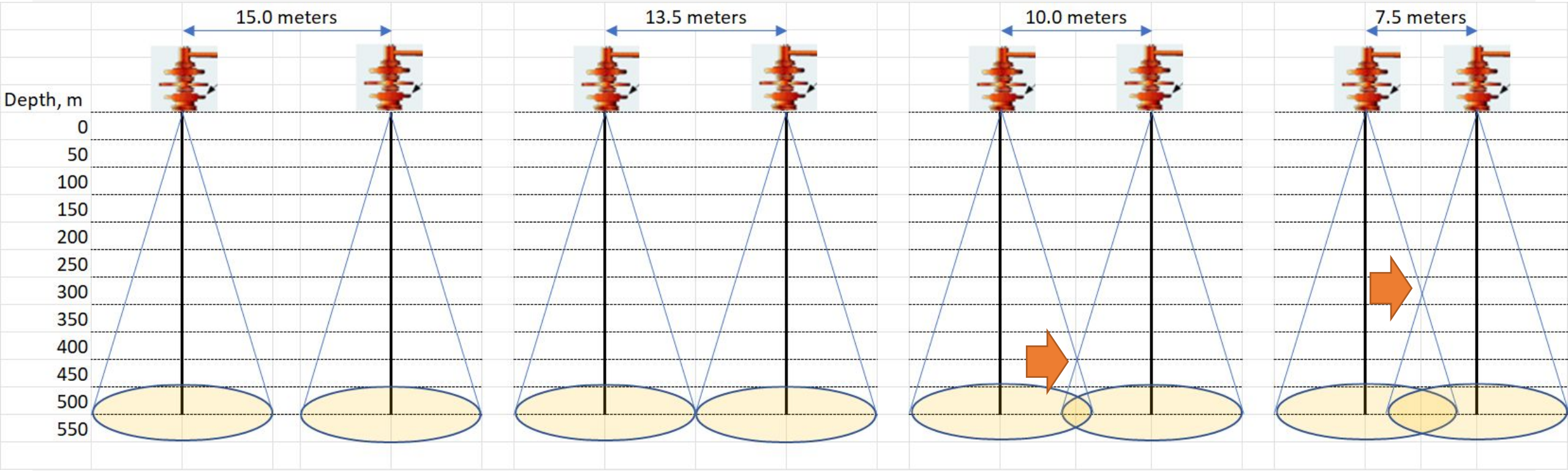
- Rig Floor Size
- Post Drilling Activity

Rig Picture: Side View for Determining Distance between Cellars

- When Rig Skid forward for next sequence well, the drilled well could be access for any well intervention such as:
  - Wireline operation
  - Well Test Pipe Installation
- Distance between Cellar (Ideal Option): **15.0 Meters**
  - Rotary Table end Rig Floor (back side): 13.5 m
  - Half of Cellar Wide: 1.5 m



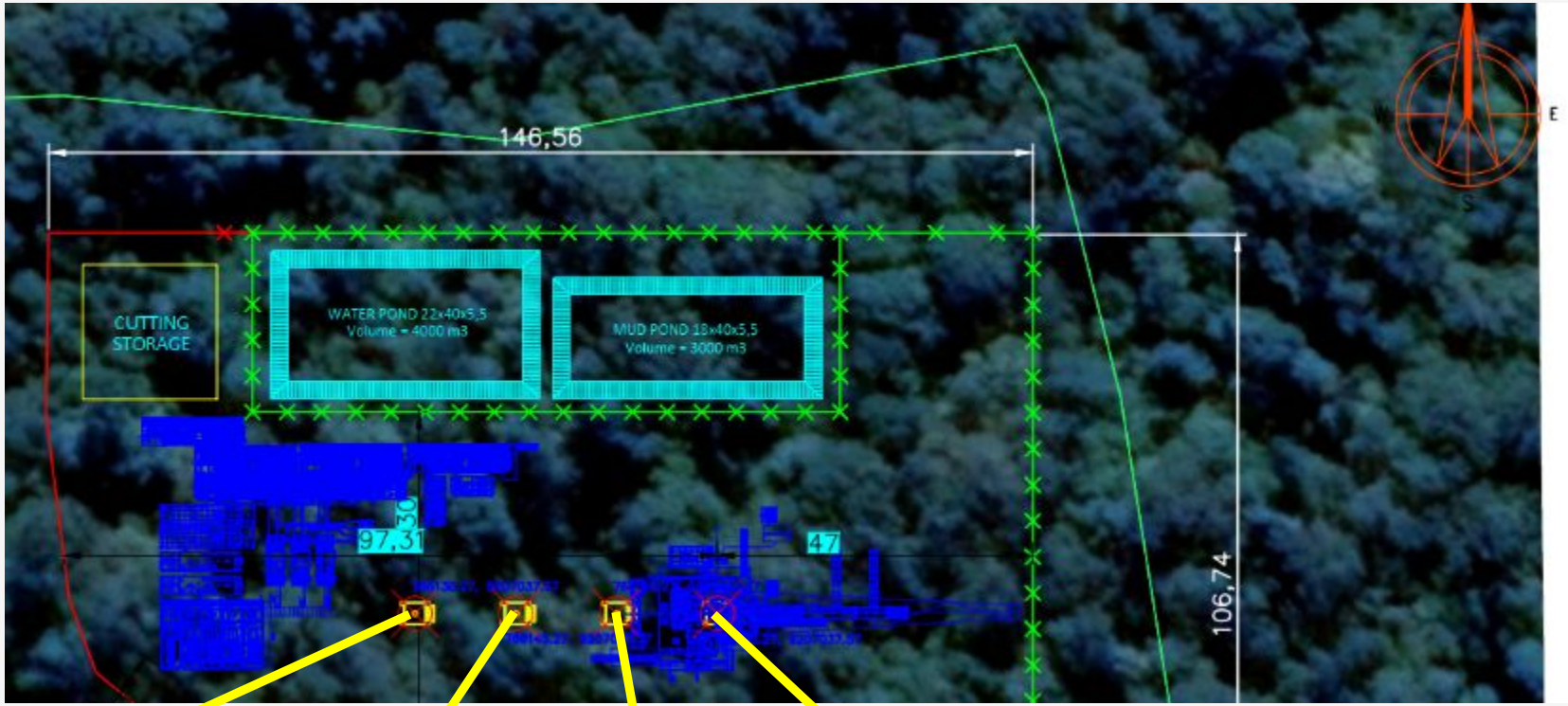
Anticollision



SF > 1 | SF = 1 | SF < 1 | SF < 1

SF = Separation Factor
SF = (Ellipse Radius-1 + Ellipse Radius-2)/ Jarak lubang

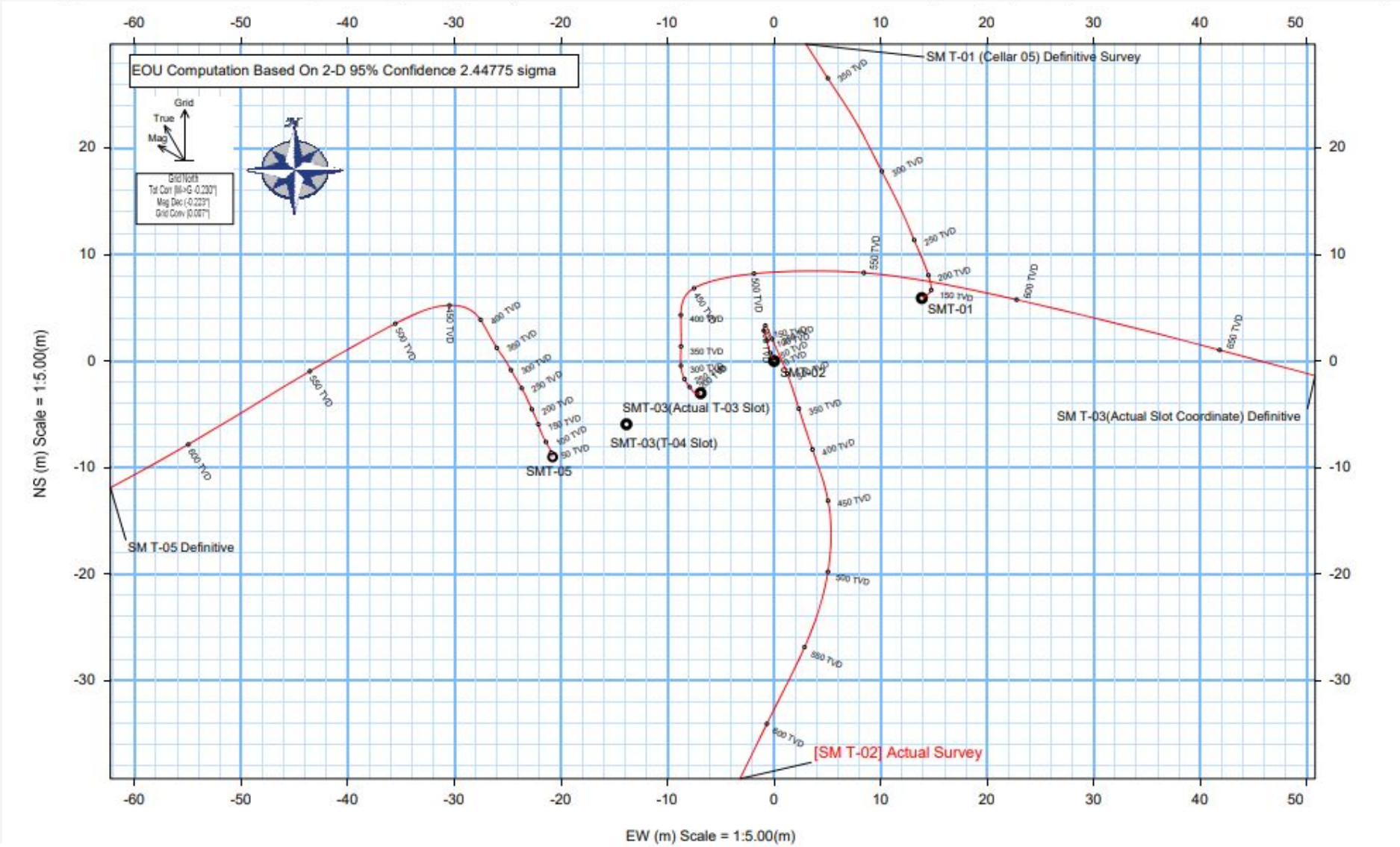
PAD BB: Cellar Assignment



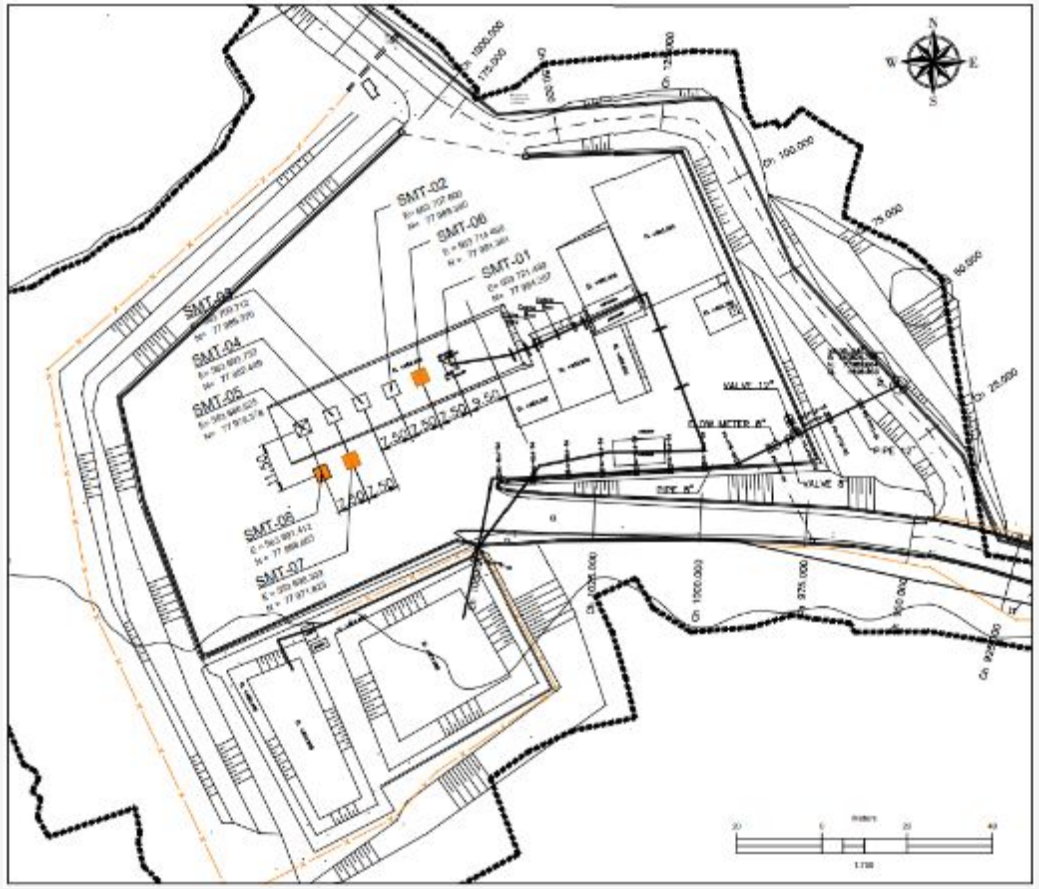
Well	Az
PTH-BB-9	215
PTH-BB-9A	168
PTH-BB-9B	245
PTH-BB-9C	135

- Cellar assignment refer to Well Trajectory
- Avoid any Crossing as possible
- Check well sequence to easy Rig Skidding
- Check Wellhead Height to avoid any obstruction during Rig Skidding

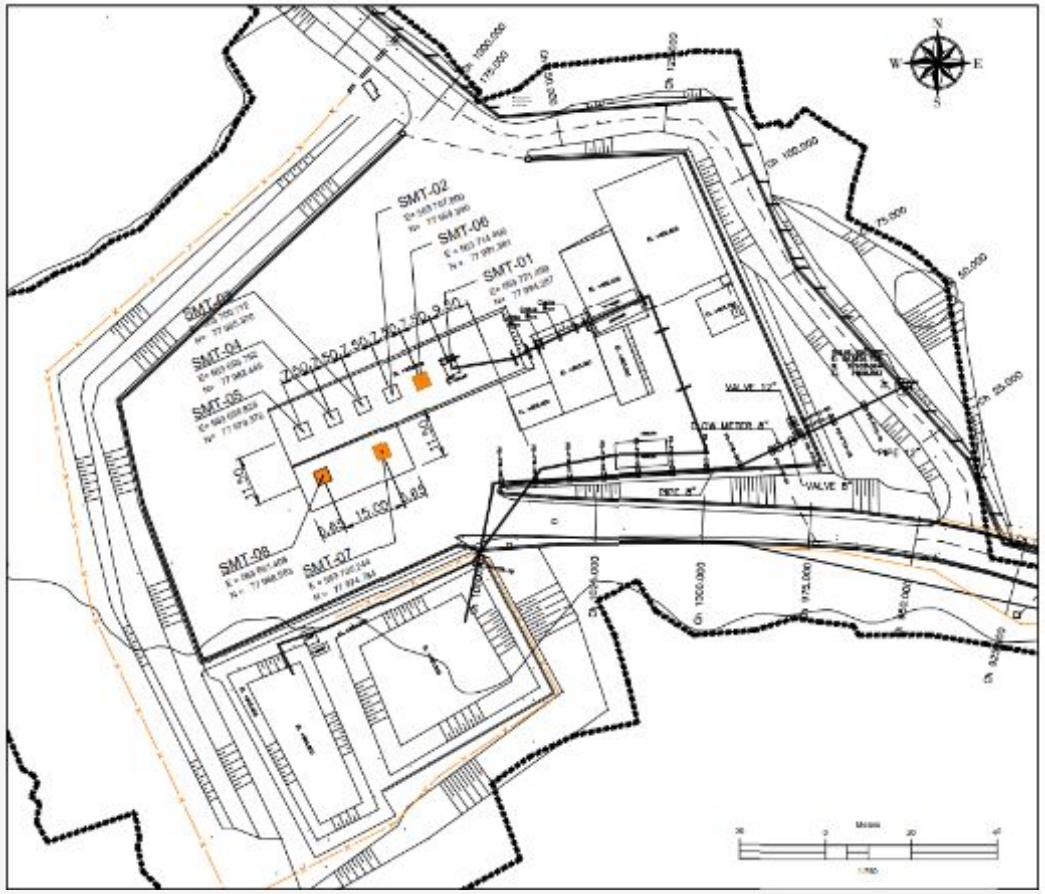
Cellar Assignment, Optimized ?



Additional Cellar to the Existing Pad. Is it safe?



Well Collision Risk – High

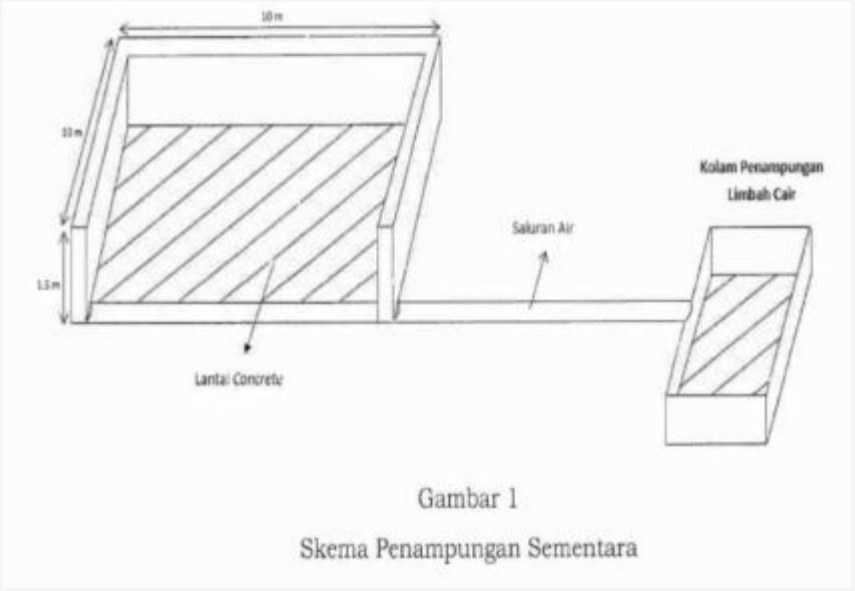


BHA for Better Control Anticollision at 26” Hole Section

BHA Configuration	Parameters	26” Hole	17-1/2” Hole	12-1/4” Hole
9-½ MM + 9-½ EM-MWD	Max GPM	1200	1200	1200
	MU Time	Longer	Longer	Longer
	Directional Control	Good	Good	Fair (*)
9-½ MM + 8-½ EM-MWD	Max GPM	1200	1200	1200
	MU Time	Fair	Fair	Fair
	Directional Control	**Poor: too limber for nudging.**	Good but Risk of Twist off	Good, MM 6-5/8” Reg
8-½ MM + 8-½ MWD	Max GPM	900	900	900
	MU Time	Faster	Faster	Faster
	Directional Control	**Not Applicable, Bit: 7-5/8” Reg Connection, GPM too low**		Good

Temporary Cutting Bunker





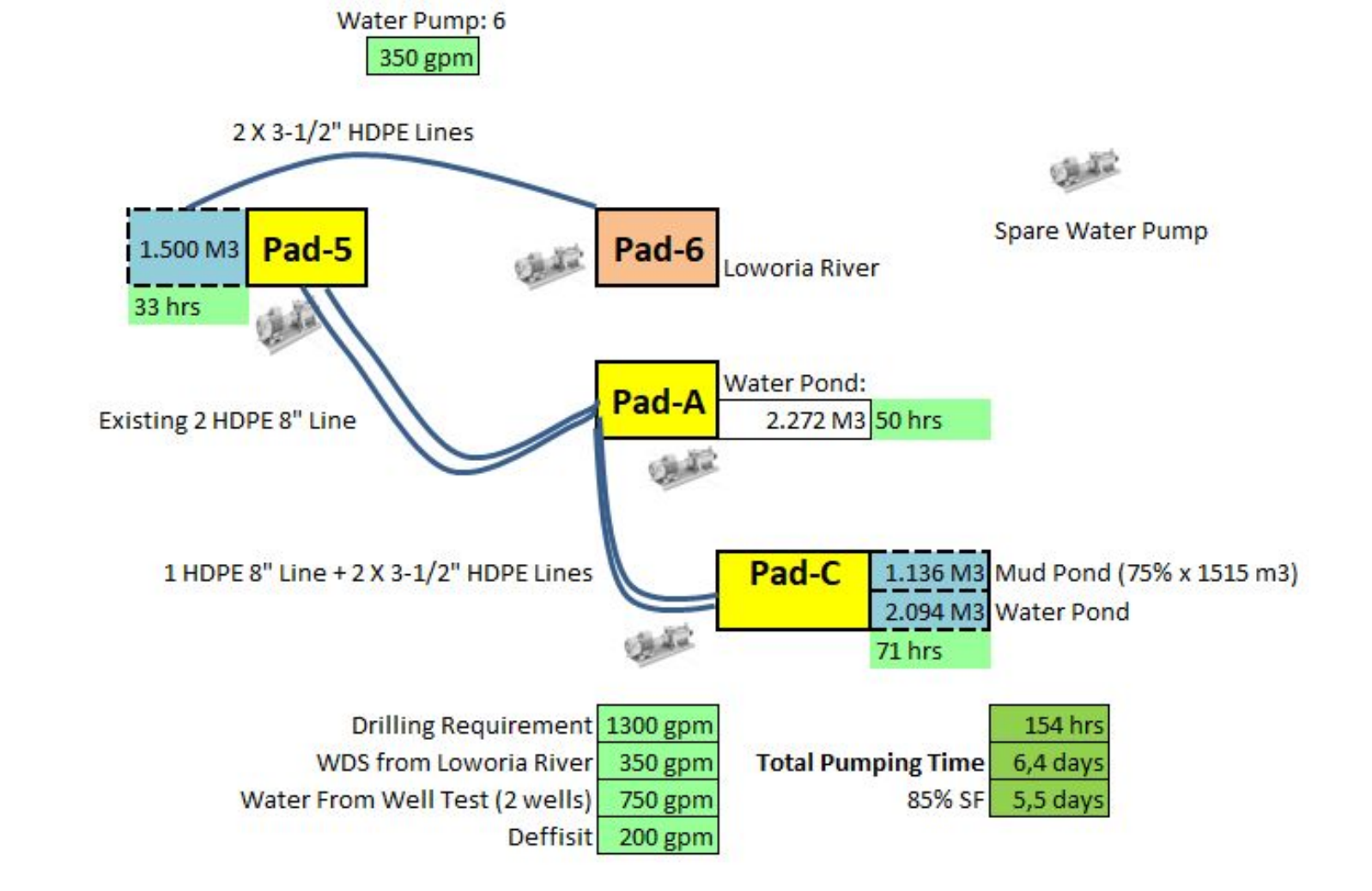
Gambar 1

Skema Penampungan Sementara

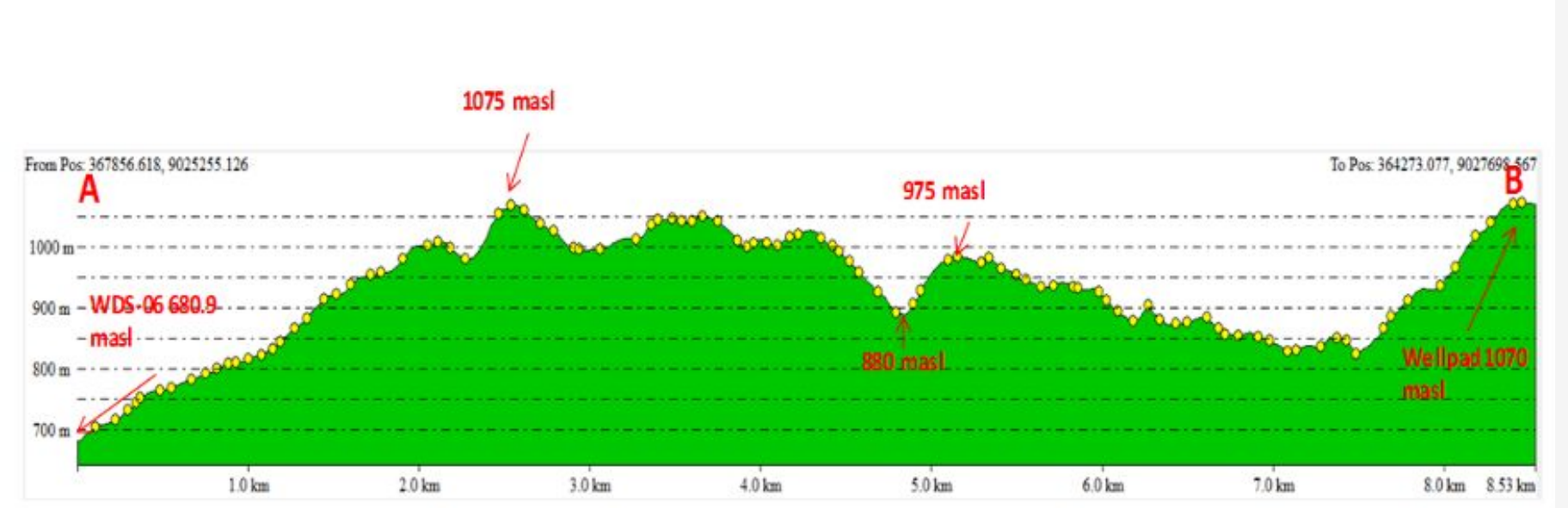
Reference: Permen ESDM No 21, tahun 2017

Water Supply Design

WATER SUPPLY – 1300 GPM

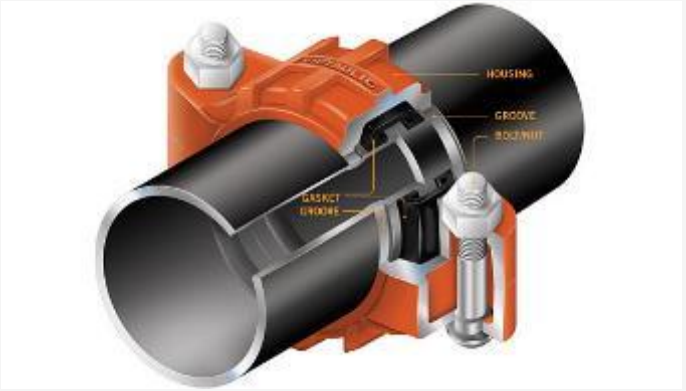


Water Supply



WATER SUPPLY – 1300 GPM

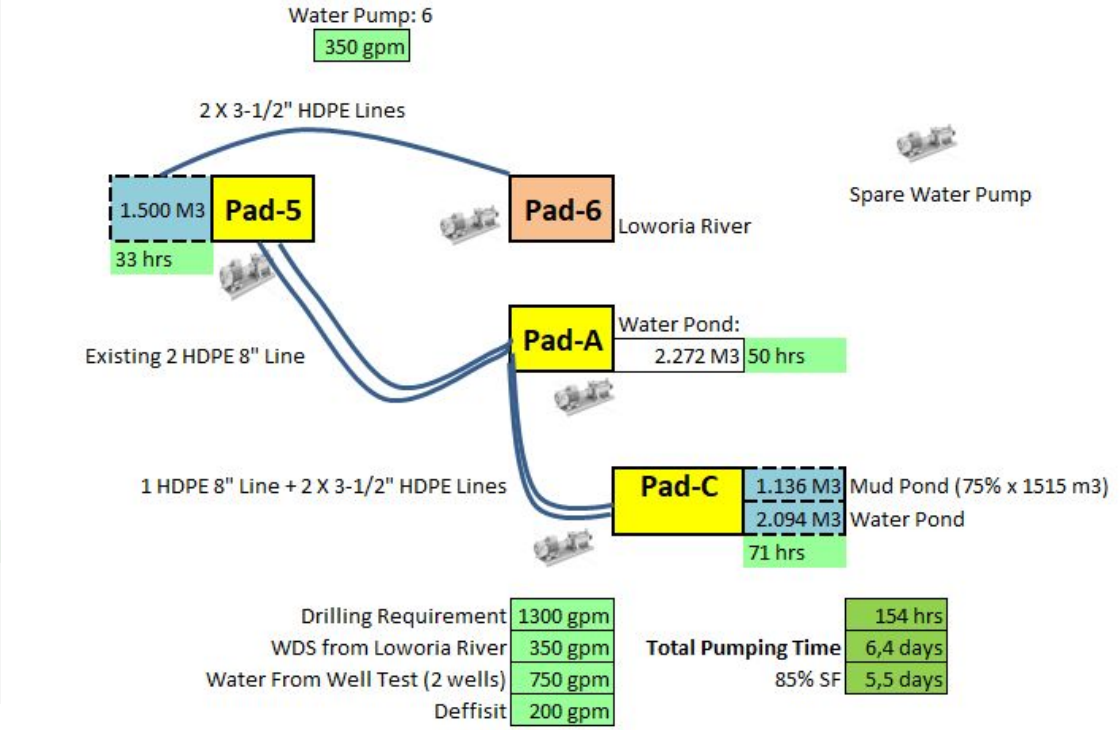
- Water Sources:
  - Water pH?
  - Water Flow rate.
  - Distance and elevation
  - Community issue





- Pipe Line Sizes:
  - 2 lines of 8”ODx6”ID HDPE pipe OR
  - 2 line of victaulic or steel pipe

- Water Pond Storage
  - 8000 – 10000 m3
  - If no water source bigger water pons up 100.000 m3



WATER SUPPLY – 1400 GPM

WATER PIPE LINE: Use Galvanize or Steel Pipe Line instead of HDPE:

- Less pressure drop due to friction
- Higher Burst Pressure capacity allow to utilize maximum water pump **FLOW RATE** and **PRESSURE capacity**.
  - **Water Pump: 600 psi**
  - **Steel Pipe: ±1000 psi**
  - **HDPE Pipe: 300 psi.**
- Less water pump requirement Less Personnel requirement Less Community issue
- Easy to repair if any pipe leak









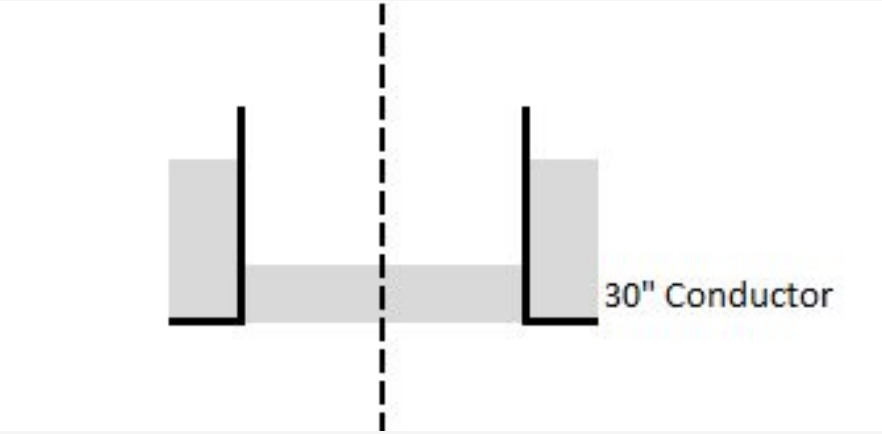


DRILLING OPERATION

CONDUCTOR INSTALLATION

Conductor Installation

- Pre-Driven by Pile Driver
- Using Auger System
- Installed by Rig: drill with 36” bit or 36” hole opener, run 30” conductor and cemented



Conductor Installation – Pre-Driven



- Diesel Pilling Hammer, Delmag D-44 or equivalent
- 30” casing with 1” thickness
- 45 Ton Crawler Crane
- Bevel machine and cutting equipment
- Welding Equipment

- Shoe Joint: 30° internal bevel.
- The Driving Operation will be stopped:
  - Reached desired Depth
  - Penetration Rate slow to 200 blows per foot.

- Application:
  - **Soft surface formation** – Oil and Gas Wellpad
  - **Not Applicable** for Geothermal, Hard surface formation

Pile Hammer – Blow/ft

Pipe Size And Wall Thickness	Blows Per ft:	Hammer Size		
		D 22	D 30	D 44
20 x .312	Normal			
	Maximum			
20 x .375	Normal	65 - 70		
	Maximum	90		
20 x .500	Normal	65 - 90	55 - 80	
	Maximum	120	110	
20 x .750	Normal	100 - 150	100 - 120	
	Maximum	160	140	
20 x 1.00	Normal	140 - 180	120 - 150	
	Maximum	200	170	
24 x .500	Normal	90 - 110	80 - 100	
	Maximum	150	140	
24 x .625	Normal	100 - 120	90 - 110	
	Maximum	170	160	
24 x .750	Normal	120 - 150	110 - 140	
	Maximum	200	180	
24 x 1.00	Normal	150 - 200	150 - 180	
	Maximum	250	200	
26 x .500	Normal	100 - 150	90 - 100	
	Maximum	200	170	
26 x .750	Normal	150 - 180	110 - 150	
	Maximum	250	200	
26 x 1.00	Normal	200 - 220	175 - 200	
	Maximum	300	250	
30 x .500	Normal	150 - 200	100 - 150	
	Maximum	250	200	
30 x .625	Normal	200 - 225	140 - 175	100 - 130
	Maximum	275	250	150
30 x .750	Normal	250 - 300	150 - 200	130 - 160
	Maximum	350	300	180
30 x 1.00	Normal	300 - 350	200 - 300	150 - 200
	Maximum	400	350	250
36 x .500	Normal	160 - 210	120 - 170	
	Maximum	260	220	
36 x .625	Normal	210 - 235		120 - 140
	Maximum	280		160
36 x .750	Normal	260 - 310	200 - 250	150 - 170
	Maximum	360	350	190
36 x 1.00	Normal	320 - 360	250 - 350	180 - 210
	Maximum	425	400	260
*48 x .750	Normal			170 - 180
	Maximum			200
*48 x 1.00	Normal			180 - 200
	Maximum			300
* With adapter				

Conductor Installation – Auger System



- 9-11 loads for equipment mobilization.
- Expensive Mobilization and Demobilization Cost
- Cost effective if we do many conductor in once time.
- Pre-installed 30” conductor (cemented) prior to rig arrival

Conductor Installation – By Rig



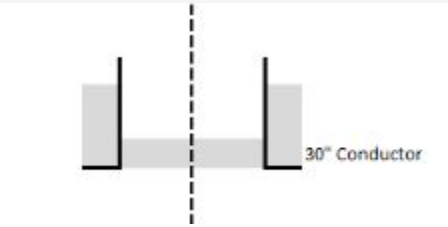


Hole Opener

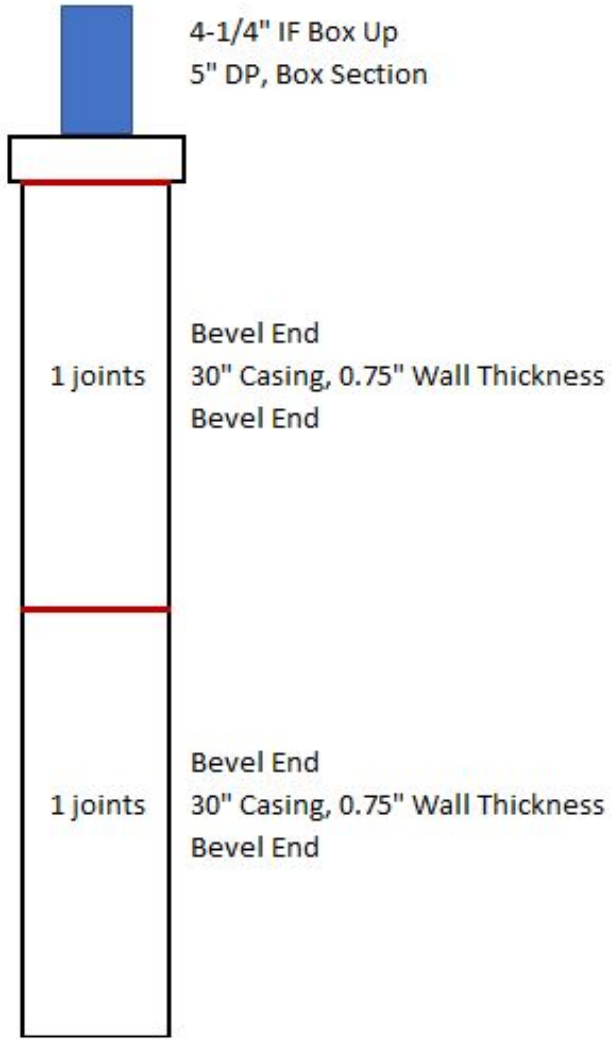


36” Mill Tooth Bit

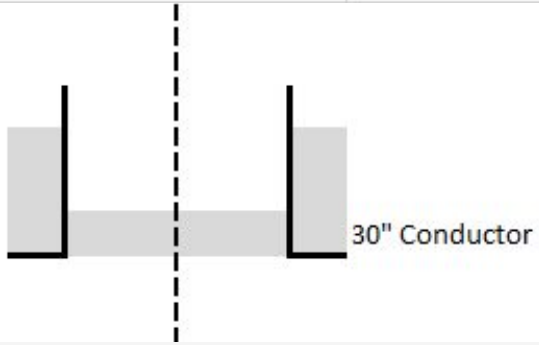
- Use 26” Bit tandem with 36” Hole Opener
- OR
- Drill with 36” Mill Tooth Bit.
- Run 30” Pipe and Cemented



Pre-Welded 2 Joints of 30” Conductors



30" CONDUCTOR INTALLATION - COST COMPARISON					
Using Auger System	Rp 7.175.000.000	14300	USD Rate	$ 501.748	Cost for 3 wells
				$ 167.249	Cost for 1 well
IF USING RIG					
Cost Description	Unit Price	Qty	Unit	Total Cost/ Well	
Rig Operation Cost	$ 80.000	1	Day	$ 80.000	
36" Hole Opener Service	$ 15.000	1	LS/well	$ 15.000	
36" Hole Opener Rental	$ 150	40	Day	$ 6.000	
36" Hole Opener Cone Replacement	$ 20.000	0,33	LS	$ 6.667	
36" Hole Opener Mob/Demob	$ 8.000	0,33	LS	$ 2.667	
Cementing	$ 250	60	bbls	$ 15.000	
				$ 125.333	
			Price escalation, 10%	$ 18.800	
			TOTAL COST TO INSTALL 30" CONDUCTOR BY RIG	**$ 144.133**	
				$ -23.116	Cost saving per well
				$ -69.348	Cost saving for 3 wells



- Do not use Rig to install 30” Conductor if hard surface formation

Other Drilling Support Area

a. Open Yard for: Casing storage, excess rig equipment, etc
b. Warehouse
c. Base camp
d. Drill Cutting Disposal
e. Temporary Staging Area
f. Access Road

Cover Warehouse for Rig Base Camps



Morning in Sokoria





DISCUSSION



Thank You!

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